Veer Narmad South Gujarat University, Surat.

Department of Information and Communication Technology

M.Sc. (Information and Communication Technology) Programme

Project Report

2nd Semester

M.Sc. (Information and Communication Technology)

2 Year Course

Year 2025 – 2026

Project Title: VerisqAI

Guided By:
Dr. Dhaval Joshi (Internal Guide)

Submitted By:
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Company Name:
Kyara Infotech

Veer Narmad South Gujarat University, Surat.

Department of Information and Communication Technology

M.Sc. (Information and Communication Technology) Programme

Certificate

This is to certify that **Mr. Krishna Parekh** with Exam Seat Number: 48 and Enrollment Number: R251100180007100043 has worked on his/her part time project work entitled **VerisqAI** – AI-Powered Vendor Risk Assessment System at **Kyara Infotech** as a partial fulfillment of the requirements for 2nd Semester - M.Sc. (Information and Communication Technology), during the **Academic Year 2025-2026**.

Date:

Place: Department of ICT, VNSGU, Surat.

Internal Project Guide	**Course Coordinator**	**Head of the Department**
M.Sc.(I.C.T.) 2nd Semester Department of I.C.T. Veer Narmad South Gujarat University, Surat	M.Sc.(I.C.T.) Programme Department of I.C.T. Veer Narmad South Gujarat University, Surat	Department of I.C.T. Veer Narmad South Gujarat University, Surat

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This is to certify that **Mr. Vignesh Chitroda** with Exam Seat Number: 20 and Enrollment Number: R25110018000710008 has worked on his/her part time project work entitled **VerisqAI** – AI-Powered Vendor Risk Assessment System at **Kyara Infotech** as a partial fulfillment of the requirements for 2nd Semester - M.Sc. (Information and Communication Technology), during the **Academic Year 2025-2026**.

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This is to certify that **Mr. Sahil Kanjariya** with Exam Seat Number: 39 and Enrollment Number: R25110018000710032 has worked on his/her part time project work entitled **VerisqAI** – AI-Powered Vendor Risk Assessment System at **Kyara Infotech** as a partial fulfillment of the requirements for 2nd Semester - M.Sc. (Information and Communication Technology), during the **Academic Year 2025-2026.**

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1. Introduction

This document presents the complete technical and functional documentation of VerisqAI, an AI-Powered Vendor Risk Assessment System developed as part of the M.Sc. (ICT) 2nd Semester project at Veer Narmad South Gujarat University, Surat. The project was developed at Kyara Infotech, Surat during the academic year 2025-2026.

1.1 Company Profile

Kyara Infotech is a software development and technology solutions company located at Palanpur Jakatnaka, Adajan, Surat, Gujarat. The company specializes in designing and developing modern software solutions that help businesses improve efficiency, automate operations, and accelerate digital transformation.

The organization provides services in Custom Software Development, Artificial Intelligence and Machine Learning Solutions, Business Process Automation, Digital Transformation, Cloud Solutions, Intelligent Chatbots, Voice AI Agents, and Microsoft Technology Platforms. The company focuses on delivering scalable, secure, and innovative technology solutions tailored to business requirements.

Kyara Infotech follows modern software engineering practices including requirement analysis, system design, application development, database management, testing, deployment, and maintenance. The company emphasizes practical implementation of emerging technologies to solve real-world business challenges.

During the development of the project "VerisqAI – AI-Powered Vendor Risk Assessment System", Kyara Infotech provided technical guidance, project mentoring, architecture consultation, and software development support to the student team. The internship provided practical exposure to full-stack web application development, database design, API development, cybersecurity concepts, software testing, and project documentation.

Address:
42, Silverpark Society,
Pal Road, Palanpur Jakatnaka,
Adajan, Surat – 395009,
Gujarat, India.

1.2 Customer Profile

1.2.1 Current System – About the Existing System

Currently, most organizations managing third-party vendor relationships rely on manual, spreadsheet-based processes to perform vendor risk assessments. Procurement teams send questionnaires to vendors via email, and responses are manually reviewed by risk analysts. The existing process involves creating questionnaires in Microsoft Word or Google Forms, emailing them to vendors, and then manually compiling the responses into Excel spreadsheets for scoring.

This manual approach is time-consuming, error-prone, and lacks consistency. There is no standardized scoring mechanism, and risk prioritization is entirely subjective, depending on the analyst reviewing the data. The absence of an AI-driven analysis layer means that subtle risk patterns across multiple vendor responses often go undetected. Audit trails are incomplete, and generating comprehensive vendor risk reports requires considerable manual effort, often taking several days per vendor.

1.2.2 Customer Detail – Working Style of the Client

The client in this case is an enterprise-level organization that manages relationships with multiple third-party vendors across different operational domains such as cloud infrastructure, software tools, logistics, and financial services. The organization has a dedicated Vendor Risk Management (VRM) team responsible for assessing and monitoring vendor compliance with internal security policies and regulatory requirements.

The VRM team uses a periodic assessment cycle where vendors are evaluated quarterly or annually based on their risk tier. Higher-risk vendors (those with access to sensitive data or critical systems) undergo more frequent and comprehensive assessments. The organization's key requirements include automated questionnaire dispatch and tracking, consistent risk scoring across all vendors, AI-powered insights to identify critical findings, and the ability to generate professional PDF reports for executive stakeholders.

2. Proposed System

2.1 Scope

VerisqAI is a web-based, AI-powered vendor risk assessment platform designed to automate and streamline the entire third-party risk management lifecycle. The system enables organizations to onboard vendors, design dynamic assessment templates, dispatch questionnaires via secure tokenized email links, collect and automatically score responses, and generate AI-driven risk insights along with comprehensive PDF reports. The platform serves two primary user types: administrators who manage the platform and oversee all vendors system-wide, and regular users (risk analysts) who manage their own vendor portfolios. The system integrates with external AI providers (OpenRouter/Gemini) to deliver intelligent risk analysis and recommendations, making it suitable for enterprise-grade vendor risk management operations.

2.2 Objective

1. To provide a centralized platform for managing end-to-end vendor risk assessment workflows, eliminating manual, spreadsheet-based processes.
2. To enable users to design flexible, reusable assessment templates with dynamic question types, scoring weights, and conditional logic.
3. To automate questionnaire dispatch to vendors via secure, tokenized email links with expiration controls.
4. To implement an automated scoring engine that calculates vendor risk scores, risk tiers (1-4), and generates structured findings based on vendor responses.
5. To integrate AI capabilities (via OpenRouter and Gemini) for generating executive summaries, identifying top risk drivers, and producing prioritized actionable recommendations.
6. To provide an administrative dashboard with system-wide analytics including user growth trends, AI usage monitoring, audit logs, and risk distribution statistics.
7. To generate professional, formatted PDF assessment reports for executive stakeholders with complete findings and AI insights.
8. To implement secure authentication using JWT tokens with OTP-based two-factor verification for all users.

2.3 Constraints

2.3.1 Hardware Constraints

- The system requires a server with a minimum of 4 GB RAM and a multi-core processor to handle concurrent API requests and AI processing.
- A stable internet connection is mandatory for AI provider integration (OpenRouter/Gemini APIs) since AI analysis is performed via external API calls.
- Client machines (user browsers) require a modern browser supporting ES2020+ JavaScript features and a minimum screen resolution of 1280x720.
- PDF generation using QuestPDF requires additional CPU resources during report compilation; dedicated server resources are recommended for production environments.

2.3.2 Software Constraints

- The backend is built exclusively on .NET 8.0 and cannot run on older .NET Framework versions or non-Microsoft runtimes.
- The database layer is tightly coupled with Microsoft SQL Server via Entity Framework Core; migration to other database engines would require significant schema modifications.
- AI-powered features depend entirely on external API availability (OpenRouter/Gemini); any downtime in these services will render AI assessment features unavailable.
- The frontend is built with React 19 using Vite; Internet Explorer and legacy browsers are not supported.
- Email dispatch functionality requires a valid SMTP server configuration; without it, questionnaire tokens cannot be sent to vendors.

2.4 Advantages

- Eliminates manual, error-prone spreadsheet-based vendor assessment workflows by providing a fully automated digital platform.
- Consistent, weighted scoring ensures that all vendors are evaluated against the same standardized criteria regardless of which analyst processes the data.
- AI-generated executive summaries, risk drivers, and prioritized recommendations save analyst time and improve the quality of risk reporting.

- Tokenized questionnaire links with expiry controls ensure that vendors can only access valid, active assessments, enhancing security.
- The template builder allows organizations to create domain-specific assessment templates (e.g., cloud security, financial compliance, data privacy) and reuse them across multiple vendors.
- Role-based access control (Admin vs. User) ensures that system-wide data remains accessible only to authorized administrators.
- Comprehensive audit logging tracks all critical system actions, providing a complete audit trail for compliance and governance purposes.
- Real-time dashboard analytics provide visibility into vendor portfolio risk distribution, assessment completion rates, and AI processing statistics.

2.5 Limitations

- The system currently does not support multi-language questionnaires; all assessment content is presented in English only.
- There is no native mobile application; the platform is accessible only through web browsers, which may affect usability on small-screen devices.
- Bulk vendor import via CSV or Excel files is not yet implemented; vendors must be added individually through the interface.
- The AI analysis relies on external LLM APIs and is subject to rate limits and token usage costs associated with those providers.
- Vendor self-registration is not supported; all vendors must be manually added by the risk analyst before a questionnaire can be dispatched.
- The breach alert module, while present in the interface, is currently a static information page and does not yet integrate with live threat intelligence feeds.

3. Environment Specification

3.1 Hardware & Software Requirements

Hardware Requirements:

Component	Specification
Processor	Intel Core i5 (8th Gen) or equivalent, Minimum 2.0 GHz
RAM	Minimum 8 GB (16 GB recommended for production)
Storage	Minimum 50 GB SSD for OS, application, and database
Network	Broadband internet connection (minimum 10 Mbps) for AI API access
Display	1280 x 720 resolution or higher
Operating System (Server)	Windows Server 2019/2022 or Ubuntu 22.04 LTS
Operating System (Client)	Windows 10+, macOS 12+, Ubuntu 20.04+

Software Requirements:

Software	Version / Details
ASP.NET Core (Backend)	.NET 8.0 Runtime
Microsoft SQL Server	SQL Server 2019 or 2022 (Express or Standard)
Entity Framework Core	EF Core 9.x with SQL Server provider
React.js (Frontend)	React 19.x with Vite 7.x build tool
Node.js	v20 LTS (for frontend build)
QuestPDF	Community Edition (for PDF generation)
OpenRouter API / Gemini API	External AI provider for LLM-based risk analysis
SMTP Server	Any SMTP-compatible email server (e.g., Gmail SMTP, SendGrid)
Web Browser	Google Chrome 120+, Mozilla Firefox 120+, Microsoft Edge 120+
Development IDE	Microsoft Visual Studio 2022 / VS Code
API Testing	Swagger UI (integrated) / Postman

3.2 Development Description

VerisqAI follows a modern full-stack architecture with a clear separation between the backend API layer and the frontend client application. The backend is built using ASP.NET Core 8 Web API following RESTful principles. It uses Entity Framework Core as the ORM layer for database interactions with Microsoft SQL Server. The application employs ASP.NET Identity for user management and JWT (JSON Web Token) authentication for securing all API endpoints.

The frontend is built with React 19 and bundled using Vite. It uses React Router DOM for client-side routing, Recharts for data visualization, and Axios for HTTP communication with the backend API. The UI is organized into two distinct portals: the Admin Portal (for super-users) and the User Portal (for risk analysts), each with its own layout, navigation, and role-specific features.

The AI integration layer is implemented as a provider pattern with the IAiProvider interface, allowing easy switching between AI providers. The current implementation uses OpenRouter (which provides access to multiple LLMs) as the primary provider. The AI service receives structured assessment data (vendor information, question-answer pairs, scores, findings) and returns a JSON response containing an executive summary, risk drivers, and prioritized recommendations.

4. System Planning

4.1 Feasibility Study

Technical Feasibility:

VerisqAI is technically feasible as it leverages widely adopted, well-documented technologies. ASP.NET Core 8 is a mature, high-performance framework with extensive community support. React 19 is the industry-leading frontend framework for building interactive UIs. Entity Framework Core simplifies database operations. OpenRouter and Gemini provide readily accessible AI APIs with comprehensive documentation. The integration of these technologies is straightforward and well within the team's capability.

Operational Feasibility:

The system is operationally feasible as it addresses a real, well-defined business problem faced by enterprise risk management teams. The user interface is designed to be intuitive, minimizing the learning curve for risk analysts. The automated workflows (questionnaire dispatch, scoring, AI analysis) replace time-consuming manual tasks, making adoption beneficial for end-users from day one.

Economic Feasibility:

The system is economically feasible. The backend framework (.NET 8), database (SQL Server Express for development), and frontend stack (React/Vite) are either free or low-cost. The primary recurring cost involves AI API usage, which scales with the number of vendor assessments processed. For small to medium enterprises, the cost savings from eliminating manual analyst hours significantly outweigh the AI API costs.

4.2 Software Engineering Model

VerisqAI was developed using the Agile (Iterative) software development methodology, structured into clearly defined phases. Development proceeded through six incremental phases: Phase 1 – Core infrastructure, authentication, and vendor management; Phase 2A – Template builder and questionnaire engine; Phase 2B – Template preview; Phase 3 – Dynamic questionnaire runtime for vendors; Phase 4 – Automated scoring engine; Phase 5 – AI assessment engine; and Phase 6 – Report generation. Each phase delivered a working, testable increment, and subsequent phases were planned based on the outcomes of earlier ones.

4.3 Risk Analysis

Risk	Impact	Probability	Mitigation
AI API unavailability or rate limits	High – AI features non-functional	Medium	Implement fallback to secondary provider; cache results
SMTP configuration failure	Medium – Questionnaire emails not sent	Low	Provide manual token link sharing as alternative
SQL Server connection failure	Critical – System unusable	Low	Connection pooling, retry policies, database backups
JWT token interception	High – Unauthorized access	Low	HTTPS enforced, short token expiry, OTP 2FA
Questionnaire token abuse	Medium – Unauthorized submissions	Low	One-time-use tokens with expiry dates
Scope creep	Medium – Delivery delays	Medium	Strict phase-based planning, backlog management

4.4 Project Schedule

4.4.1 Task Dependency

The project tasks follow a sequential dependency chain where each phase depends on the completion of the prior phase. Database schema design and authentication (Phase 1) is the foundation upon which all subsequent modules depend. The template builder (Phase 2A) must be complete before the questionnaire runtime (Phase 3) can be developed, since the runtime reads template definitions. The scoring engine (Phase 4) depends on completed questionnaire responses. The AI engine (Phase 5) depends on scoring data. PDF report generation (Phase 6) depends on both scoring and AI data.

4.4.2 Timeline Chart (Gantt Overview)

Phase / Module	Start	End	Duration	Status
Phase 1: Auth, Vendor Management, DB Design	Jan 2026	Jan 2026	3 weeks	Completed
Phase 2A: Template Builder, Questionnaire Engine	Jan 2026	Feb 2026	3 weeks	Completed
Phase 2B: Template Preview Mode	Feb 2026	March 2026	2 week	Planned
Phase 3: Vendor Questionnaire Runtime	March 2026	March 2026	2 weeks	Completed
Phase 4: Dynamic Scoring Engine	March 2026	April 2026	3 weeks	Completed
Phase 5: AI Assessment Engine	April 2026	April 2026	2 week	Completed
Phase 6: PDF Report Generation	May 2026	May 2026	1 week	Completed
Admin Dashboard & Monitoring	May 2026	May 2026	2 week	Completed
Testing & Bug Fixing	May 2026	Jun 2026	1 week	In Progress

4.4.3 Project Table

No	Task	Owner	Dependency
1	Database Schema Design & EF Core Migrations	Backend Developer	None
2	JWT Authentication & ASP.NET Identity Setup	Backend Developer	Task 1
3	OTP / TOTP Two-Factor Authentication	Backend Developer	Task 2
4	Vendor CRUD Operations & Dashboard API	Backend Developer	Task 2
5	Assessment Template Builder API	Backend Developer	Task 4
6	Template Builder UI (React)	Frontend Developer	Task 5
7	Questionnaire Dispatch & Token System	Full Stack	Task 5
8	Vendor Questionnaire Form (Public)	Frontend Developer	Task 7
9	Dynamic Scoring Engine	Backend Developer	Task 8
10	AI Assessment Service (OpenRouter/Gemini)	Backend Developer	Task 9
11	AI Intelligence UI (Frontend)	Frontend Developer	Task 10
12	PDF Report Generation (QuestPDF)	Backend Developer	Task 9,10
13	Admin Dashboard & Monitoring	Full Stack	Task 4,10
14	Audit Logging System	Backend Developer	Task 2
15	Testing & Quality Assurance	QA / All	All Tasks

5. System Analysis

5.1 Detailed SRS (Software Requirements Specification)

Module 1: Authentication & User Management

Description: This module handles all user registration, login, and session management operations. It implements JWT-based authentication combined with OTP (One-Time Password) email verification for two-factor authentication.

Input: User email, password, OTP code

Events: User registration, login attempt, OTP verification, token refresh

Output: JWT access token, user role, authentication status

Validations: Email format validation, password strength check, OTP expiry check, duplicate email check

Constraints: OTP expires after 10 minutes; JWT tokens have configurable expiry

Module 2: Vendor Management

Description: This module enables risk analysts to maintain their vendor portfolio. Each vendor is associated with the authenticated user and tracks assessment status, risk tier, and questionnaire state.

Input: Vendor name, domain, contact email, risk tier

Events: Add vendor, view vendor details, update vendor status, delete vendor

Output: Vendor list with risk scores, questionnaire statuses, scorecard history

Validations: Vendor name and domain required; email must be valid format

Constraints: Vendors are user-scoped; users can only view/manage their own vendors

Module 3: Assessment Template Builder

Description: This module provides a drag-and-drop-style interface for creating reusable assessment templates. Templates consist of sections, and each section contains questions with configurable types, scoring weights, severity levels, and answer options.

Input: Template name, description, sections, questions, question options, weights, question type (boolean, multiple-choice, text, scale)

Events: Create template, add section, add question, reorder sections/questions, import section from library

Output: Saved template with complete section/question hierarchy, template version

Validations: Template name required; questions must have at least one option for choice-type questions; weight values must be numeric

Constraints: Templates are user-owned; users cannot modify other users' templates

Module 4: Questionnaire Dispatch & Vendor Response

Description: This module manages the lifecycle of a questionnaire from dispatch to completion. A questionnaire is created by linking a vendor to an assessment template, generating a unique tokenized link, and emailing it to the vendor's contact. The vendor accesses a public form via the token and submits responses.

Input: Vendor ID, template ID, contact email, questionnaire responses (question - answer pairs)

Events: Send questionnaire, vendor opens link, vendor submits responses, decline questionnaire, cancel questionnaire

Output: Questionnaire record with status (Sent/Pending/Completed/Declined/Cancelled), stored responses

Validations: Token must be valid and non-expired; required questions must be answered before submission

Constraints: Each token is single-use per questionnaire session; questionnaires expire after a configurable period

Module 5: Scoring Engine & Risk Assessment

Description: Upon questionnaire submission, the scoring engine evaluates vendor responses against the template's predefined preferred answers, weights, and severity levels. It calculates a composite risk score, assigns a risk tier (1-4), and identifies findings.

Input: Completed questionnaire responses, assessment template with scoring configuration

Events: Automated trigger on questionnaire completion

Output: Scorecard with risk score (0-100), risk tier (1-4), findings list with severity levels (Critical, High, Medium, Low)

Validations: All required questions must have responses before scoring proceeds

Constraints: Scoring is deterministic based on template configuration; cannot be manually overridden

Module 6: AI Assessment Engine

Description: This module sends structured assessment data to an external LLM (via OpenRouter) to generate AI-powered insights. The AI returns an executive summary, list of top risk drivers, and prioritized recommendations with rationale.

Input: Vendor details, scorecard data, findings, questionnaire Q&A pairs

Events: User triggers AI analysis from vendor details page

Output: Executive summary text, risk drivers array, recommendations (title, description, priority, category, rationale), confidence score

Validations: AI response must be valid JSON matching the expected schema; malformed responses are rejected

Constraints: Dependent on external AI API availability; results are stored in database to prevent repeated API calls

Module 7: PDF Report Generation

Description: This module generates a comprehensive, professionally formatted PDF assessment report using QuestPDF. The report includes vendor overview, risk scores, findings table, AI executive summary, risk drivers, and prioritized recommendations.

Input: Scorecard ID, vendor data, findings, AI insights

Events: User clicks download report button on vendor details page

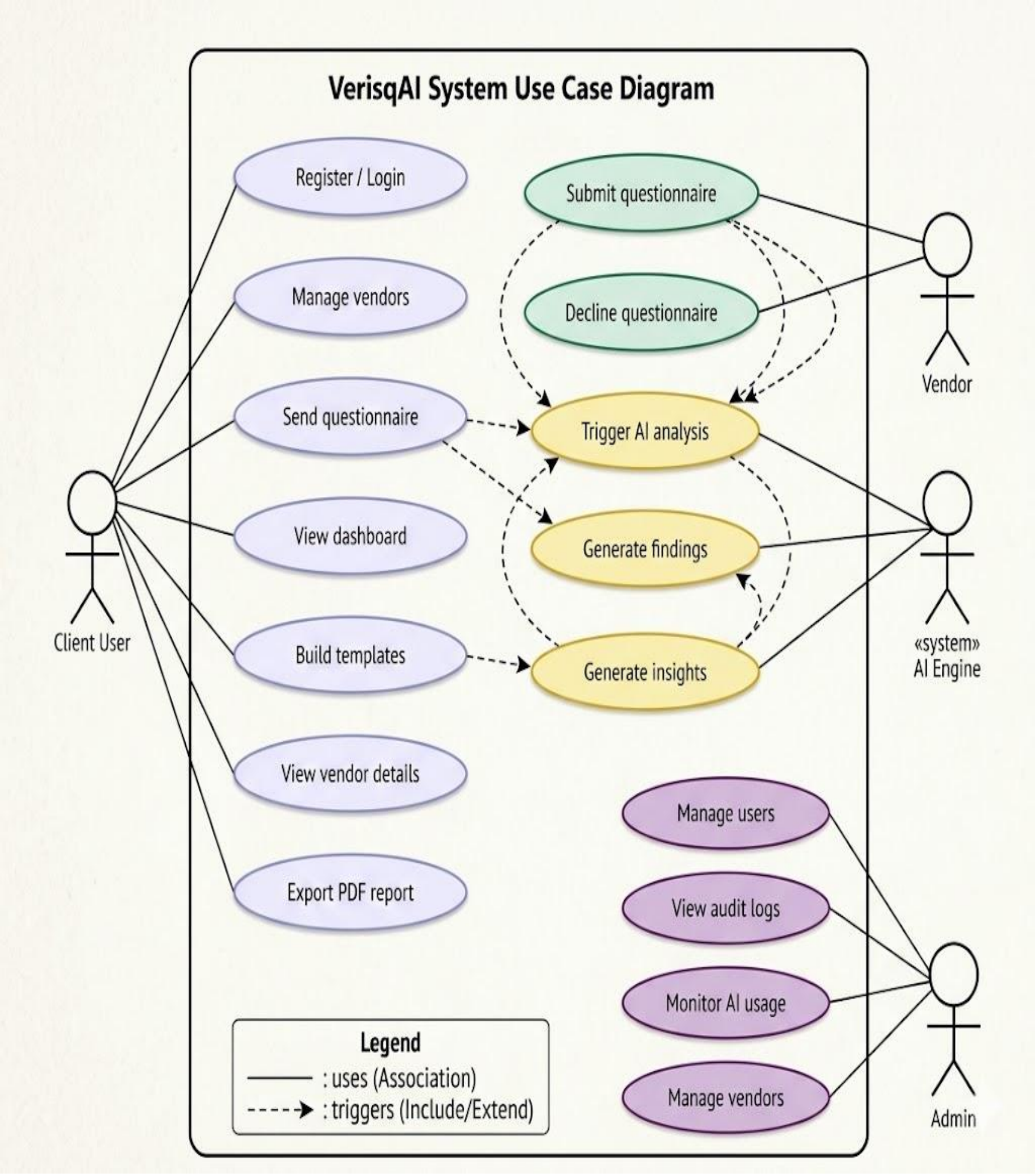
Output: Downloadable PDF file containing full vendor risk assessment report

Constraints: PDF generation requires completed scorecard; unavailable if no assessment has been processed

5.2 UML Diagrams

5.2.1 Use Case Diagram

The primary actors in VerisqAI are: Administrator, Risk Analyst (User), and Vendor. The Administrator can manage all users, view all vendors system-wide, monitor AI activity, review audit logs, and access the global admin dashboard. The Risk Analyst (User) can manage their own vendors, design assessment templates, send questionnaires to vendors, view vendor scorecards and AI insights, and download PDF reports. The Vendor (external actor) can access the questionnaire via a tokenized link, fill out and submit the assessment form, or decline the questionnaire with a reason.



5.2.2 CRC (Class Responsibility Collaborator)

ApplicationUser

Responsibilities	Collaborators
• Register & authenticate	• OtpToken
• Own vendors & templates	• Vendor
• Track status (Pending/Active)	• AssessmentTemplate
• Manage profile info	• AuditLog

Vendor

Responsibilities	Collaborators
• Store vendor identity & domain	• ApplicationUser
• Track risk score & risk tier	• Scorecard
• Maintain questionnaire status	• Questionnaire
• Link scorecard history	

Questionnaire

Responsibilities	Collaborators
• Issue secure token to vendor	• Vendor
• Track lifecycle (Sent→Done)	• AssessmentTemplate
• Record decline reason	• QuestionnaireResponse
• Enforce expiry date	• Scorecard

AssessmentTemplate

Responsibilities	Collaborators
• Define template structure	• ApplicationUser
• Version & activate templates	• AssessmentSection
• Belong to a user	• Questionnaire
• Provide section hierarchy	

AssessmentSection

Responsibilities	Collaborators
• Group questions by topic	• AssessmentTemplate
• Control display order	• AssessmentQuestion
• Describe section purpose	

AssessmentQuestion

Responsibilities	Collaborators
• Define question type & weight	• AssessmentSection
• Store severity & category	• AssessmentQuestionOption
• Support conditional logic	• QuestionnaireResponse
• Enforce required/optional	

Scorecard

Responsibilities	Collaborators
• Calculate risk score & tier	• Vendor
• Track processing status	• Questionnaire
• Hold findings collection	• Finding
• Link AI output	• AiAssessmentInsight

Finding

Responsibilities	Collaborators
• Store finding title & description	• Scorecard
• Classify severity (Critical–Low)	• AiAssessmentService
• Timestamp creation	

AiAssessmentInsight

Responsibilities	Collaborators
• Store executive summary	• Scorecard
• Store risk drivers (JSON)	• IAiProvider
• Track confidence & model	• AiExecutionAudit

AuditLog

Responsibilities	Collaborators
• Record system events	• ApplicationUser
• Classify severity & source	• All domain entities
• Identify entity & actor	

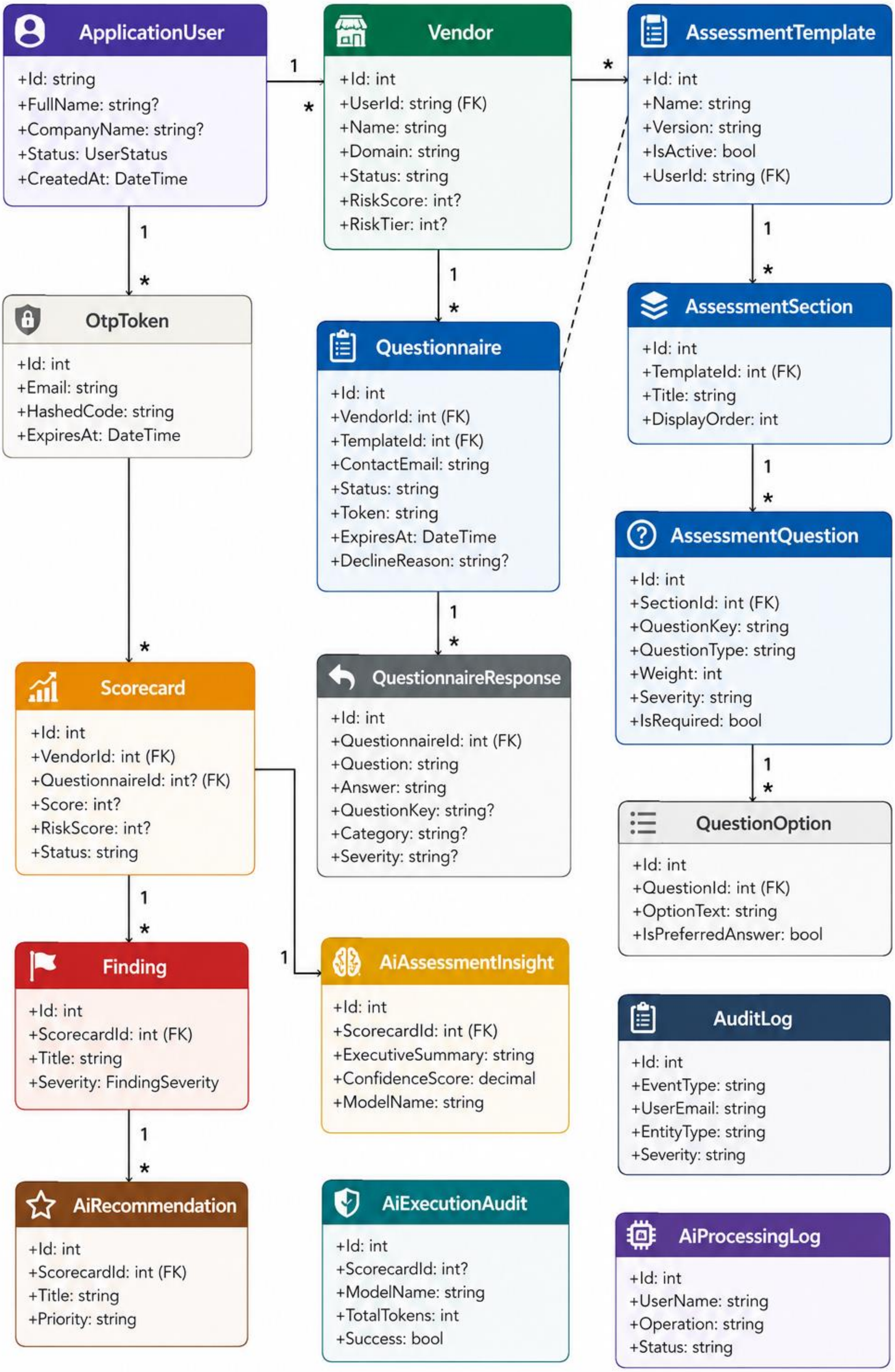
AiRecommendation

Responsibilities	Collaborators
• Store priority & category	• Scorecard
• Provide actionable rationale	

OtpToken

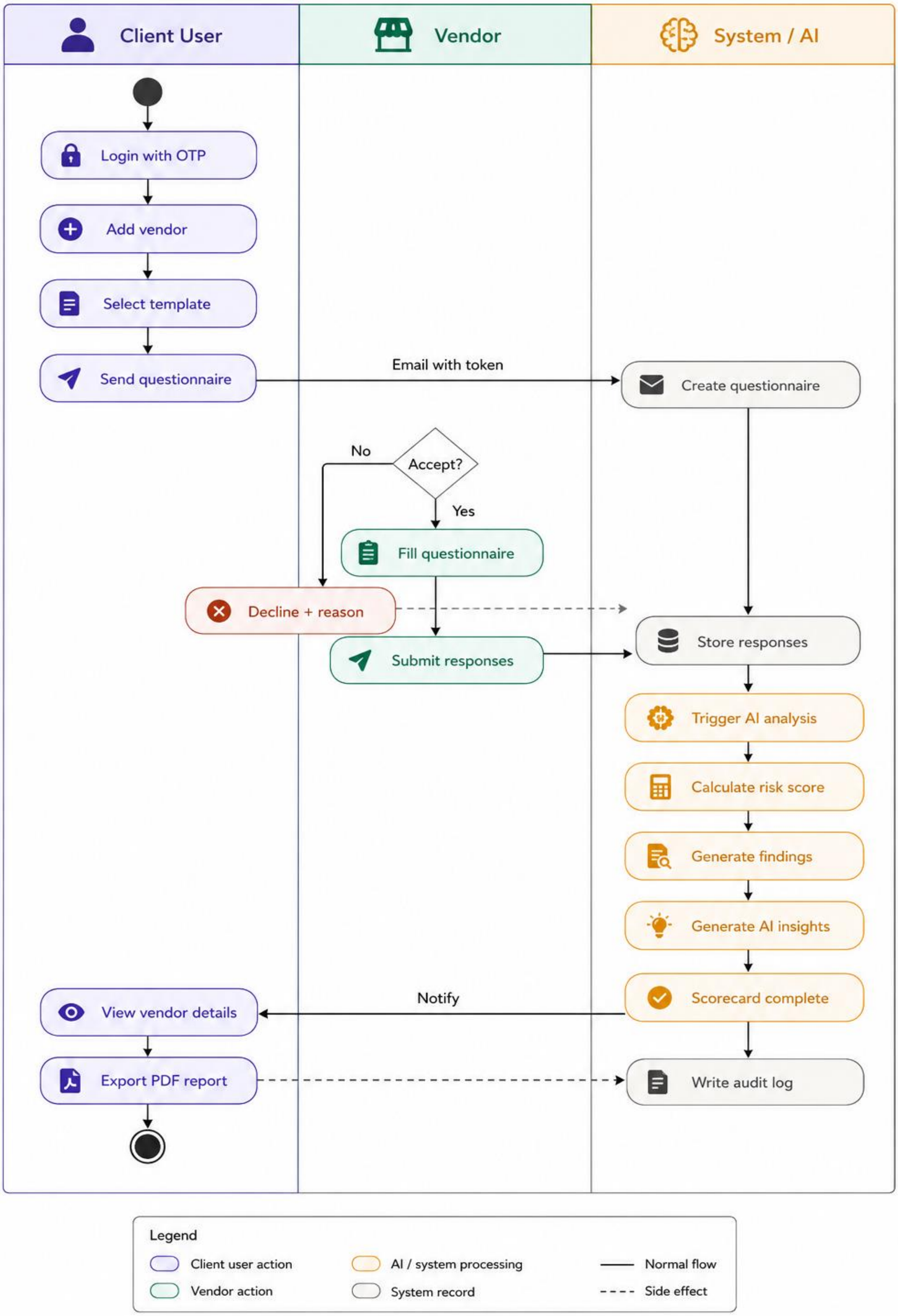
Responsibilities	Collaborators
• Hash & verify OTP code	• ApplicationUser
• Track lockout & expiry	

5.2.3 Class Diagram (Key Entities & Relationships)

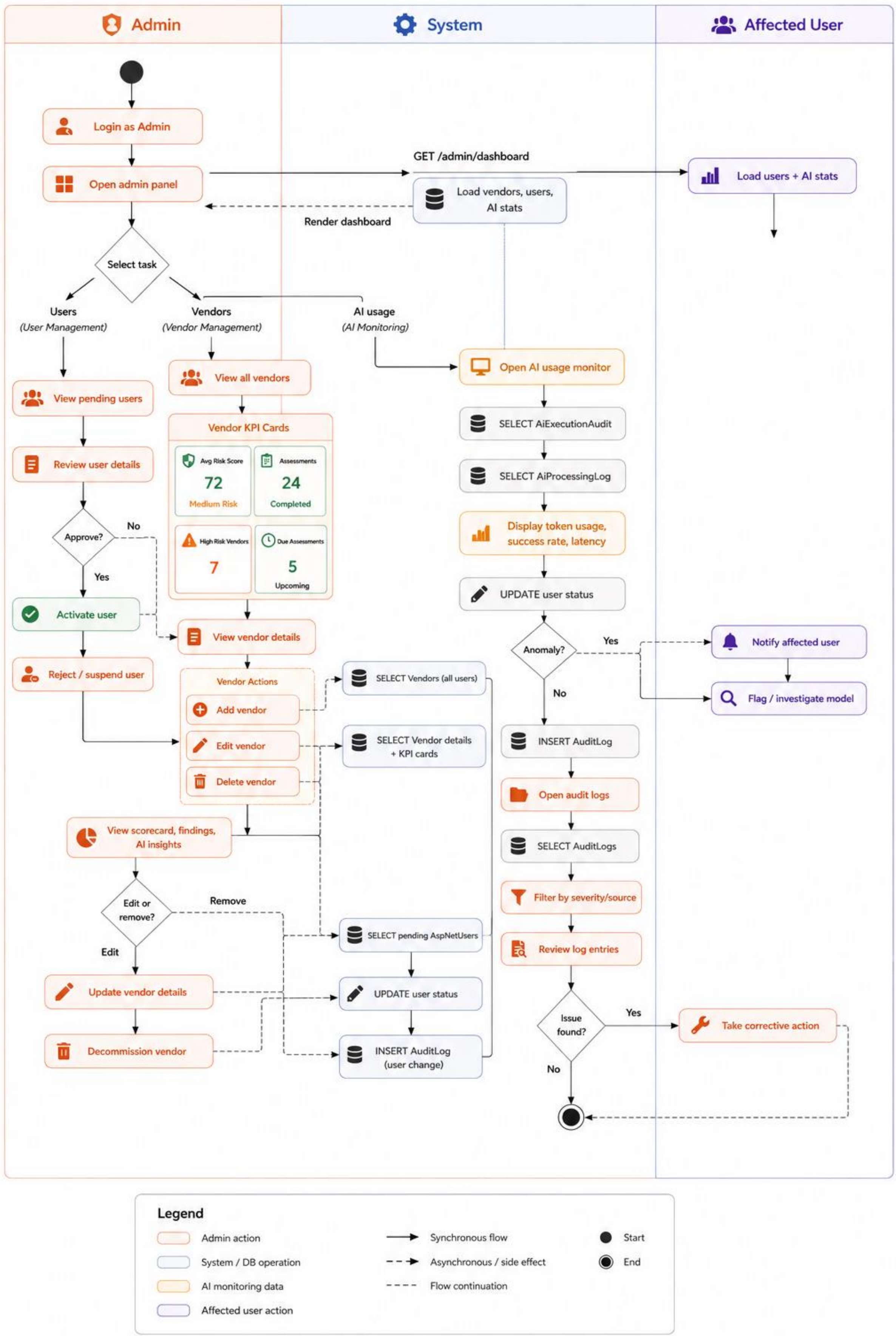


5.2.4 Activity Diagram

5.2.4.1 User Side - Activity Diagram

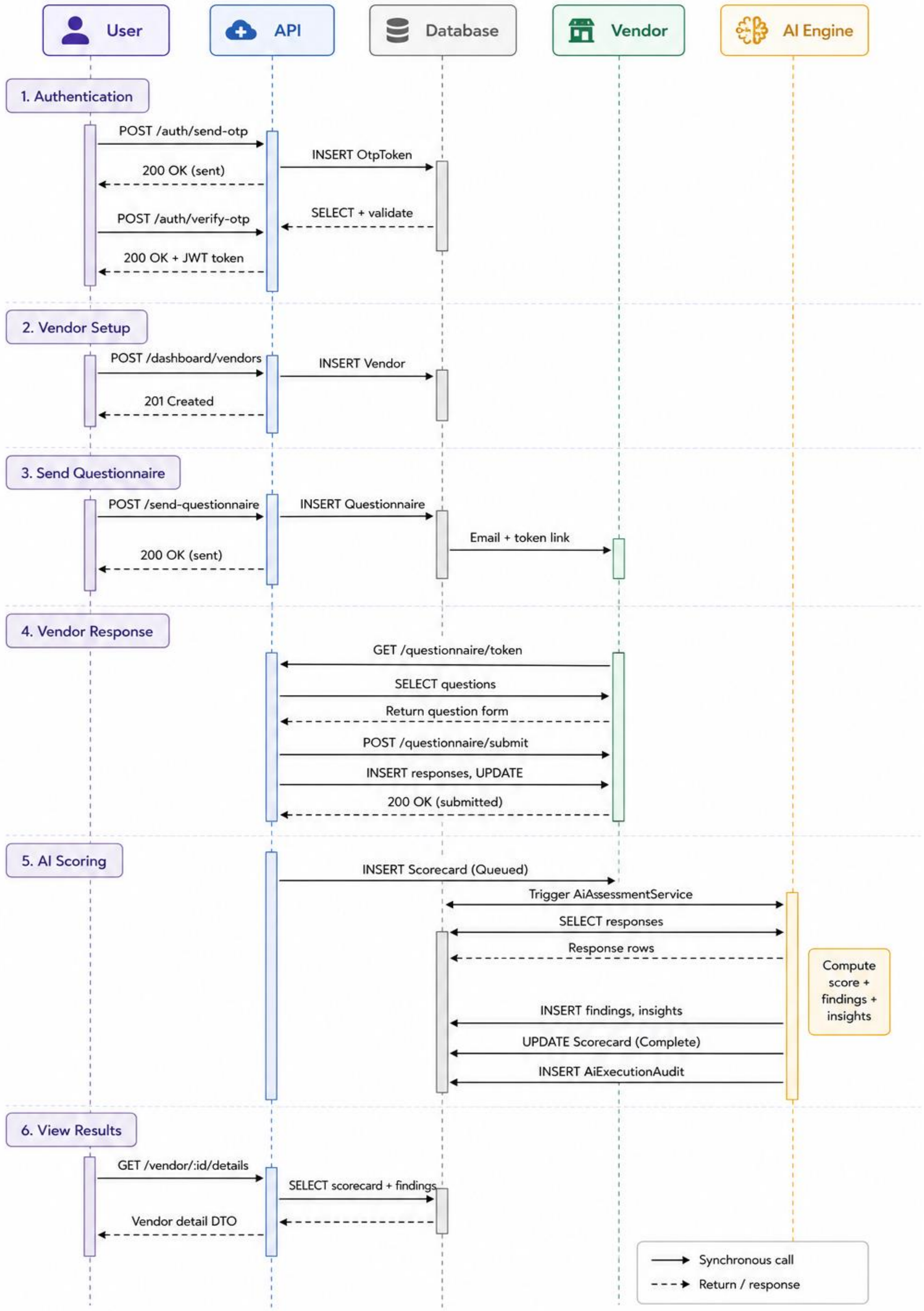


5.2.4.2 Admin Side - Activity Diagram

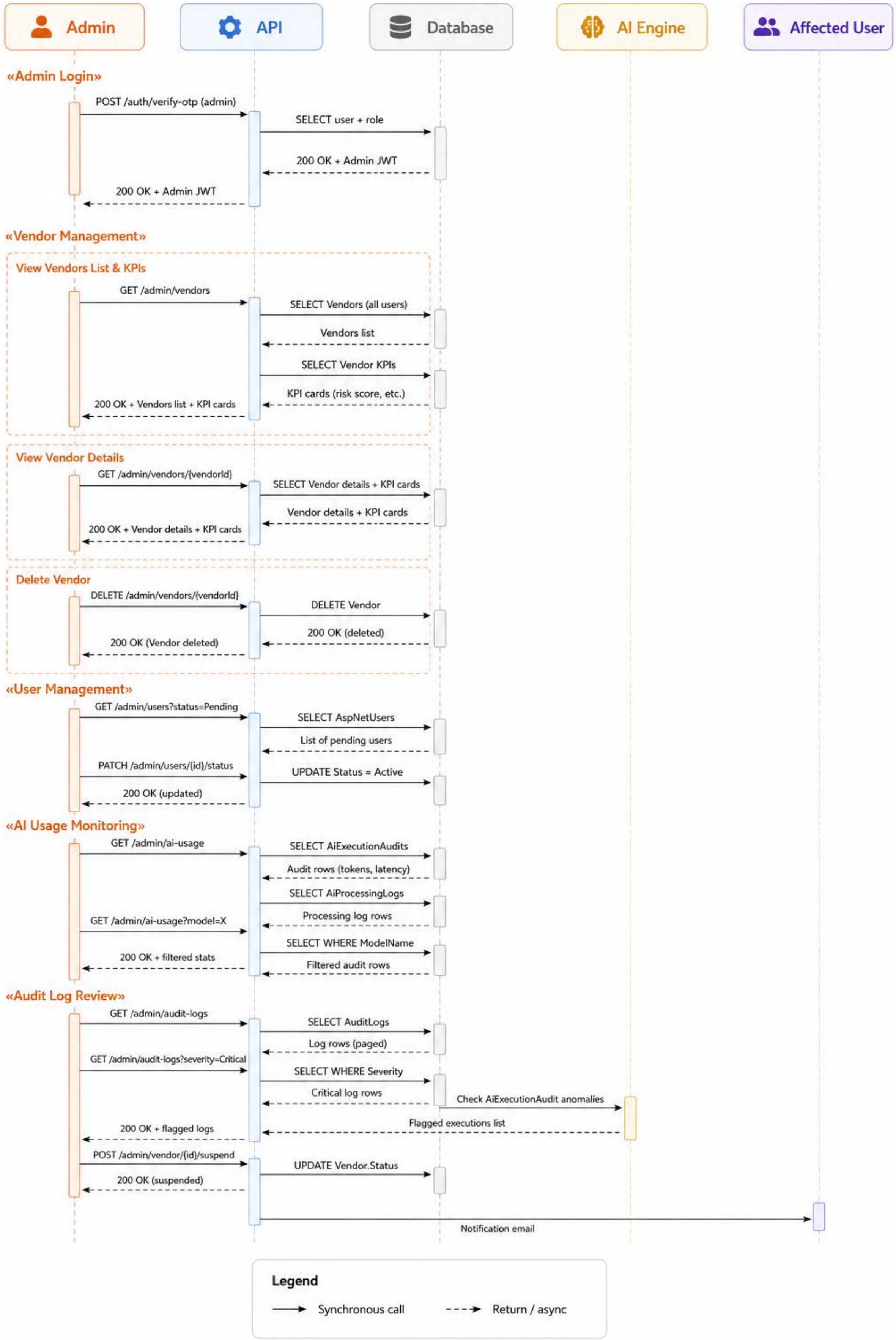


5.2.5 Sequence Diagram

5.2.4.1 User Side - Activity Diagram

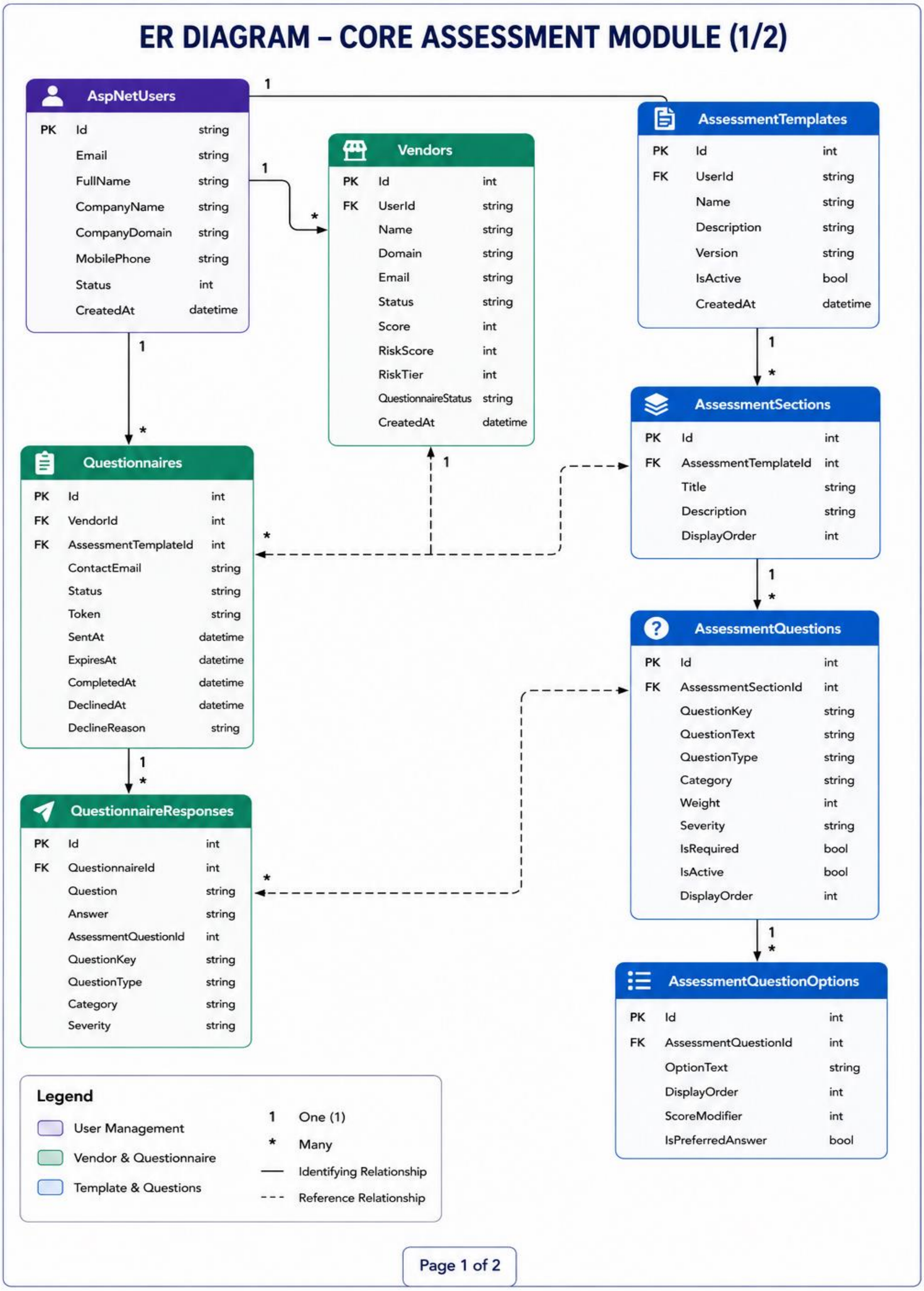


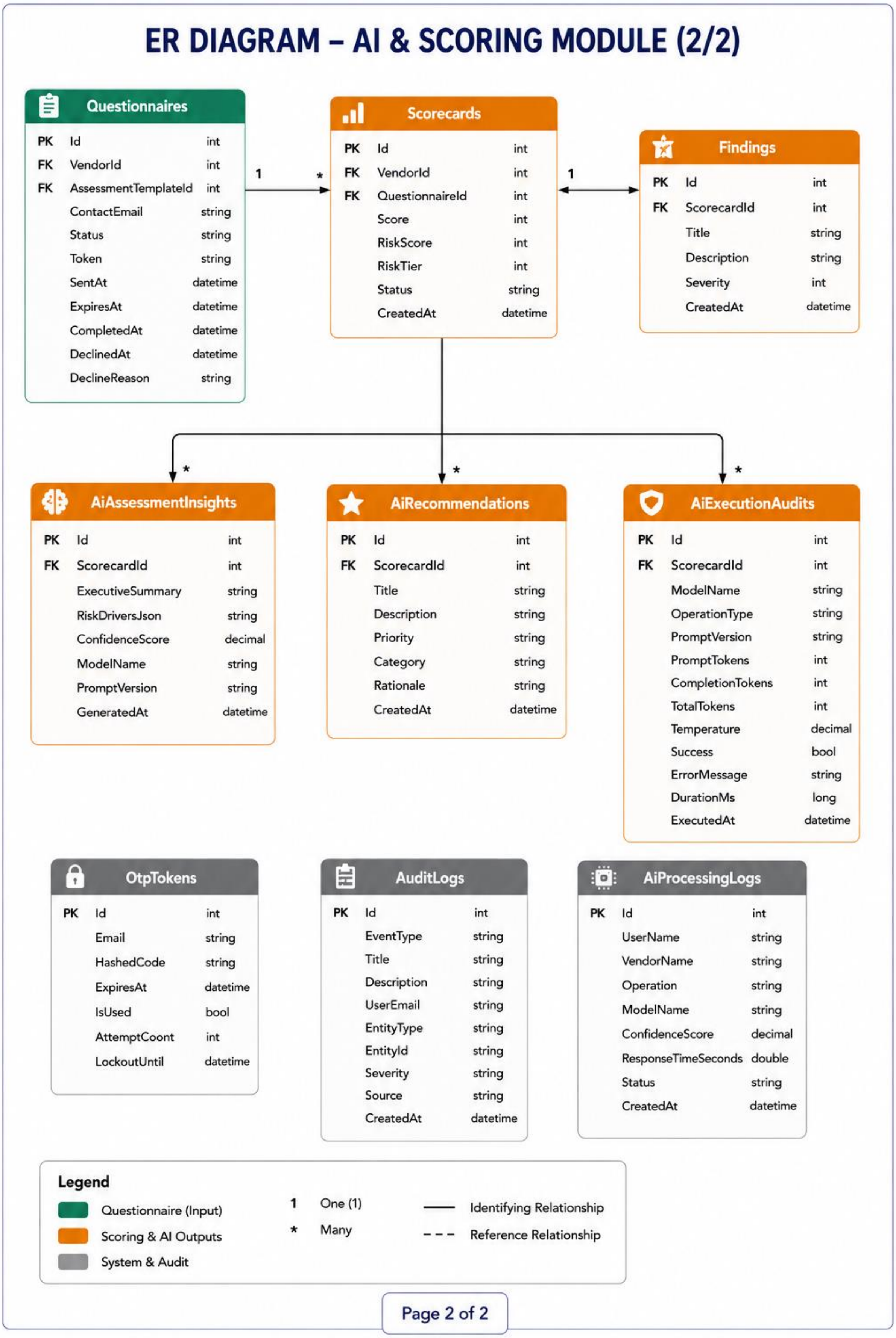
5.2.4.2 Admin Side - Activity Diagram



5.3 E-R Diagram

The Entity-Relationship model of VerisqAI encompasses the following key entities and their relationships:




ER DIAGRAM – AI & SCORING MODULE (2/2)
Questionnaires
PK Id int
FK VendorId int
FK AssessmentTemplateId int
ContactEmail string
Status string
Token string
SentAt datetime
ExpiresAt datetime
CompletedAt datetime
DeclinedAt datetime
DeclineReason string
1
*
Scorecards
PK Id int
FK VendorId int
FK QuestionnaireId int
Score int
RiskScore int
RiskTier int
Status string
CreatedAt datetime
1
Findings
PK Id int
FK ScorecardId int
Title string
Description string
Severity int
CreatedAt datetime
*
AiAssessmentInsights
PK Id int
FK ScorecardId int
ExecutiveSummary string
RiskDriversJson string
ConfidenceScore decimal
ModelName string
PromptVersion string
GeneratedAt datetime
*
AiRecommendations
PK Id int
FK ScorecardId int
Title string
Description string
Priority string
Category string
Rationale string
CreatedAt datetime
*
AiExecutionAudits
PK Id int
FK ScorecardId int
ModelName string
OperationType string
PromptVersion string
PromptTokens int
CompletionTokens int
TotalTokens int
Temperature decimal
Success bool
ErrorMessage string
DurationMs long
ExecutedAt datetime
OtpTokens
PK Id int
Email string
HashedCode string
ExpiresAt datetime
IsUsed bool
AttemptCoont int
LockoutUntil datetime
AuditLogs
PK Id int
EventType string
Title string
Description string
UserEmail string
EntityType string
EntityId string
Severity string
Source string
CreatedAt datetime
AiProcessingLogs
PK Id int
UserName string
VendorName string
Operation string
ModelName string
ConfidenceScore decimal
ResponseTimeSeconds double
Status string
CreatedAt datetime
Legend
Questionnaire (Input)
Scoring & AI Outputs
System & Audit
1 One (1)
* Many
Identifying Relationship
Reference Relationship
Page 2 of 2

6. Software Design

6.1 Database Design

AiAssessmentInsights

Column Name	Data Type	Nullable	Description
Id	int	No	Unique identifier for the AI assessment insight record
ScorecardId	int	No	References the associated scorecard
ExecutiveSummary	nvarchar(max)	Yes	AI-generated executive summary of assessment results
RiskDriversJson	nvarchar(3000)	Yes	JSON representation of key risk drivers identified by AI
ConfidenceScore	decimal(5,4)	No	Confidence level of the AI-generated assessment
ModelName	nvarchar(100)	No	Name of the AI model used for analysis
PromptVersion	nvarchar(50)	No	Version of the prompt used for AI generation
GeneratedAt	datetime2(7)	No	Date and time when the AI insight was generated

AiExecutionAudits

Column Name	Data Type	Nullable	Description
Id	int	No	Unique identifier for the AI execution audit record
ScorecardId	int	No	References the associated scorecard being analyzed
ModelName	nvarchar(100)	No	Name of the AI model used for execution
OperationType	nvarchar(100)	No	Type of AI operation performed (e.g., Insight Generation, Recommendation Generation)
PromptVersion	nvarchar(100)	No	Version of the prompt template used
PromptTokens	int	No	Number of input tokens sent to the AI model

CompletionTokens	int	No	Number of output tokens generated by the AI model
TotalTokens	int	No	Total token count consumed during execution
Temperature	decimal(3,2)	No	Temperature setting used for AI generation
Success	bit	No	Indicates whether the AI execution completed successfully
ErrorMessage	nvarchar(max)	Yes	Error details if execution failed
ExecutionTimeMs	int	Yes	Total execution time in milliseconds
CreatedAt	datetime2(7)	No	Date and time when the execution occurred

AiProcessingLogs

Column Name	Data Type	Nullable	Description
Id	int	No	Unique identifier for the processing log record
UserName	nvarchar(200)	No	Name of the user who initiated the AI operation
VendorName	nvarchar(200)	No	Name of the vendor being analyzed
Operation	nvarchar(200)	No	Type of AI operation performed
ModelName	nvarchar(100)	No	Name of the AI model used
ConfidenceScore	decimal(18,2)	Yes	Confidence score returned by the AI model
ResponseTimeSeconds	float	Yes	Time taken by the AI model to generate the response
Status	nvarchar(50)	No	Processing status (e.g., Success, Failed, In Progress)
CreatedAt	datetime2(7)	No	Date and time when the processing activity occurred

AiRecommendations

Column Name	Data Type	Nullable	Description
Id	int	No	Unique identifier for the recommendation record
ScorecardId	int	No	References the associated scorecard
Title	nvarchar(200)	No	Short title of the recommendation
Description	nvarchar(max)	No	Detailed recommendation provided by the AI model
Priority	nvarchar(50)	No	Priority level of the recommendation (High, Medium, Low)
Category	nvarchar(100)	No	Recommendation category (e.g., Security, Compliance, Governance)
Rationale	nvarchar(max)	No	AI-generated reasoning explaining why the recommendation was created
CreatedAt	datetime2(7)	No	Date and time when the recommendation was generated

AspNetRoleClaims

Column Name	Data Type	Nullable	Description
Id	int	No	Unique identifier for the role claim record
RoleId	nvarchar(450)	No	References the associated role
ClaimType	nvarchar(max)	Yes	Type of claim assigned to the role
ClaimValue	nvarchar(max)	Yes	Value of the claim assigned to the role

AspNetRoles

Column Name	Data Type	Nullable	Description
Id	nvarchar(450)	No	Unique identifier for the role
Name	nvarchar(256)	Yes	Role name used within the application
NormalizedName	nvarchar(256)	Yes	Normalized version of the role name used for lookups and comparisons
ConcurrencyStamp	nvarchar(max)	Yes	Value used for optimistic concurrency checking during updates

AspNetUserClaims

Column Name	Data Type	Nullable	Description
Id	int	No	Unique identifier for the user claim record
UserId	nvarchar(450)	No	References the associated user
ClaimType	nvarchar(max)	Yes	Type of claim assigned to the user
ClaimValue	nvarchar(max)	Yes	Value of the claim assigned to the user

AspNetUserLogins

Column Name	Data Type	Nullable	Description
LoginProvider	nvarchar(128)	No	Name of the external authentication provider
ProviderKey	nvarchar(128)	No	Unique identifier provided by the external authentication provider
ProviderDisplayName	nvarchar(max)	Yes	Friendly display name of the authentication provider
UserId	nvarchar(450)	No	References the associated user account

AspNetUserRoles

Column Name	Data Type	Nullable	Description
UserId	nvarchar(450)	No	References the user account
RoleId	nvarchar(450)	No	References the assigned role

AspNetUsers

Column Name	Data Type	Nullable	Description
Id	nvarchar(450)	No	Unique identifier for the user
FullName	nvarchar(200)	Yes	Full name of the user
CompanyName	nvarchar(200)	Yes	Organization or company associated with the user
CompanyDomain	nvarchar(200)	Yes	Company website or domain name
MobilePhone	nvarchar(20)	Yes	User's mobile contact number
Status	nvarchar(50)	Yes	Current account status (Active, Inactive, Pending, etc.)
UserName	nvarchar(256)	Yes	Username used for login
NormalizedUserName	nvarchar(256)	Yes	Normalized username for searching and comparisons
Email	nvarchar(256)	Yes	User email address
NormalizedEmail	nvarchar(256)	Yes	Normalized email address
EmailConfirmed	bit	No	Indicates whether the user's email has been verified
PasswordHash	nvarchar(max)	Yes	Encrypted password hash
SecurityStamp	nvarchar(max)	Yes	Security token used to invalidate authentication sessions
ConcurrencyStamp	nvarchar(max)	Yes	Value used for optimistic concurrency control
PhoneNumber	nvarchar(max)	Yes	User's phone number

PhoneNumberConfirmed	bit	No	Indicates whether the phone number has been verified
TwoFactorEnabled	bit	No	Indicates whether two-factor authentication is enabled
LockoutEnd	datetimeoffset(7)	Yes	Date and time when account lockout expires
LockoutEnabled	bit	No	Indicates whether account lockout is enabled
AccessFailedCount	int	No	Number of failed login attempts
CreatedAt	datetime2(7)	No	Date and time when the user account was created

AspNetUserTokens

Column Name	Data Type	Nullable	Description
UserId	nvarchar(450)	No	References the associated user account
LoginProvider	nvarchar(128)	No	Authentication provider associated with the token
Name	nvarchar(128)	No	Name or type of the token
Value	nvarchar(max)	Yes	Token value or security credential

AssessmentQuestionOptions

Column Name	Data Type	Nullable	Description
Id	int	No	Unique identifier for the question option
AssessmentQuestionId	int	No	References the associated assessment question
OptionText	nvarchar(500)	No	Text displayed as the answer option
DisplayOrder	int	No	Determines the display sequence of options
ScoreModifier	int	No	Score adjustment applied when the option is selected
IsPreferredAnswer	bit	No	Indicates whether the option represents the preferred or recommended answer

AssessmentQuestions

Column Name	Data Type	Nullable	Description
Id	int	No	Unique identifier for the assessment question
AssessmentSectionId	int	No	References the section to which the question belongs
QuestionKey	nvarchar(150)	No	Unique key used to identify the question
QuestionText	nvarchar(2000)	No	The actual question displayed to the vendor
QuestionType	nvarchar(50)	No	Type of question (Text, Radio, Checkbox, Dropdown, Yes/No, etc.)
Category	nvarchar(100)	Yes	Classification category of the question
Weight	int	No	Weight assigned for scoring calculations
Severity	nvarchar(50)	No	Risk severity level associated with the question
IsRequired	bit	No	Indicates whether answering the question is mandatory
IsActive	bit	No	Indicates whether the question is currently active
DisplayOrder	int	No	Determines the display sequence within the section
DependsOnQuestionKey	nvarchar(150)	Yes	Question key on which this question depends
DependsOnValue	nvarchar(500)	Yes	Value required to display this question

AssessmentSections

Column Name	Data Type	Nullable	Description
Id	int	No	Unique identifier for the assessment section
AssessmentTemplateId	int	No	References the assessment template to which the section belongs
Title	nvarchar(200)	No	Name of the assessment section
Description	nvarchar(1000)	Yes	Detailed description of the section
DisplayOrder	int	No	Determines the display sequence of sections within the assessment template

AssessmentTemplates

Column Name	Data Type	Nullable	Description
Id	int	No	Unique identifier for the assessment template
Name	nvarchar(200)	No	Name of the assessment template
Description	nvarchar(1000)	Yes	Detailed description of the assessment template
Version	int	No	Version number of the assessment template
IsActive	bit	No	Indicates whether the template is active and available for use
CreatedAt	datetime2(7)	No	Date and time when the template was created
UserId	nvarchar(450)	No	References the user who created the template

AuditLogs

Column Name	Data Type	Nullable	Description
Id	int	No	Unique identifier for the audit log record
EventType	nvarchar(100)	No	Type of event being recorded (Create, Update, Delete, Login, Logout, etc.)
Title	nvarchar(200)	No	Short title describing the event
Description	nvarchar(1000)	No	Detailed description of the event activity
UserEmail	nvarchar(200)	No	Email address of the user who performed the action
EntityType	nvarchar(200)	No	Name of the entity affected by the action
EntityId	nvarchar(200)	No	Identifier of the affected entity
Severity	nvarchar(100)	No	Severity level of the event (Info, Warning, Error, Critical)
Source	nvarchar(100)	No	Source module or component that generated the event
CreatedAt	datetime2(7)	No	Date and time when the event occurred

Findings

Column Name	Data Type	Nullable	Description
Id	int	No	Unique identifier for the finding record
ScorecardId	int	No	References the associated vendor assessment scorecard
Title	nvarchar(200)	No	Short title describing the finding
Description	nvarchar(max)	No	Detailed explanation of the identified issue or risk
Severity	int	No	Severity level assigned to the finding
CreatedAt	datetime2(7)	No	Date and time when the finding was generated

OtpTokens

Column Name	Data Type	Nullable	Description
Id	int	No	Unique identifier for the OTP record
Email	nvarchar(256)	No	Email address associated with the OTP request
HashedCode	nvarchar(max)	No	Hashed version of the generated OTP code
ExpiresAt	datetime2(7)	No	Date and time when the OTP expires
IsUsed	bit	No	Indicates whether the OTP has already been used
AttemptCount	int	No	Number of verification attempts made using the OTP
CreatedAt	datetime2(7)	No	Date and time when the OTP was generated
LockoutUntil	datetime2(7)	Yes	Date and time until which verification is temporarily locked after excessive failed attempts

QuestionnaireResponses

Column Name	Data Type	Nullable	Description
Id	int	No	Unique identifier for the questionnaire response
QuestionnaireId	int	No	References the questionnaire being completed
Question	nvarchar(max)	No	Text of the assessment question
Answer	nvarchar(max)	No	Vendor's response to the question
AssessmentQuestionId	int	Yes	References the original assessment question
Category	nvarchar(100)	Yes	Category associated with the question
QuestionKey	nvarchar(150)	Yes	Unique key identifying the question
QuestionType	nvarchar(50)	Yes	Type of question (Text, Radio, Dropdown, Yes/No, etc.)
Severity	nvarchar(50)	Yes	Risk severity associated with the question

Questionnaires

Column Name	Data Type	Nullable	Description
Id	int	No	Unique identifier for the questionnaire
VendorId	int	No	References the vendor receiving the questionnaire
ContactEmail	nvarchar(max)	No	Email address where the questionnaire invitation is sent
Status	nvarchar(max)	No	Current status of the questionnaire
SentAt	datetime2(7)	No	Date and time when the questionnaire was sent
CompletedAt	datetime2(7)	Yes	Date and time when the questionnaire was completed
Token	nvarchar(max)	No	Unique access token used to securely access the questionnaire
AssessmentTemplateId	int	No	References the assessment template assigned to the questionnaire
CancelledAt	datetime2(7)	Yes	Date and time when the questionnaire was cancelled
DeclineReason	nvarchar(1000)	Yes	Reason provided when the questionnaire is declined
DeclinedAt	datetime2(7)	Yes	Date and time when the questionnaire was declined
ExpiresAt	datetime2(7)	Yes	Date and time when the questionnaire access expires
StartedAt	datetime2(7)	Yes	Date and time when the vendor begins completing the questionnaire

Scorecards

Column Name	Data Type	Nullable	Description
Id	int	No	Unique identifier for the scorecard
VendorId	int	No	References the vendor being assessed
Score	int	Yes	Overall assessment score calculated from questionnaire responses
RiskScore	int	Yes	Calculated risk score representing vendor risk exposure
RiskTier	int	Yes	Risk classification tier assigned to the vendor
Status	nvarchar(max)	No	Current assessment status (Completed, In Progress, Pending, etc.)
CreatedAt	datetime2(7)	No	Date and time when the scorecard was generated
QuestionnaireId	int	Yes	References the questionnaire used for the assessment

Vendor

Column Name	Data Type	Nullable	Description
Id	int	No	Unique identifier for the vendor
UserId	nvarchar(450)	No	References the user who created or manages the vendor
Name	nvarchar(200)	No	Name of the vendor organization
Domain	nvarchar(200)	No	Vendor website domain
Status	nvarchar(50)	No	Current status of the vendor
Score	int	Yes	Overall assessment score
RiskScore	int	Yes	Calculated vendor risk score
Findings	int	Yes	Total number of findings identified
RiskTier	int	Yes	Risk classification tier
QuestionnaireStatus	nvarchar(50)	Yes	Current questionnaire status
CreatedAt	datetime2(7)	No	Date and time when the vendor record was created
Email	nvarchar(256)	Yes	Primary contact email address of the vendor

6.2 Interface Design

The user interface of VerisqAI is developed using React.js with a component-based structure, styled using custom CSS and Tailwind-based utility classes, and uses React Router for client-side navigation. The interface is divided into three major areas: the public-facing landing/marketing pages, the authenticated user dashboard (for organisations managing vendors and assessments), and the admin panel (for platform administrators to monitor users, vendors, AI activity and audit logs). The sitemap below shows the overall navigational structure of the application.

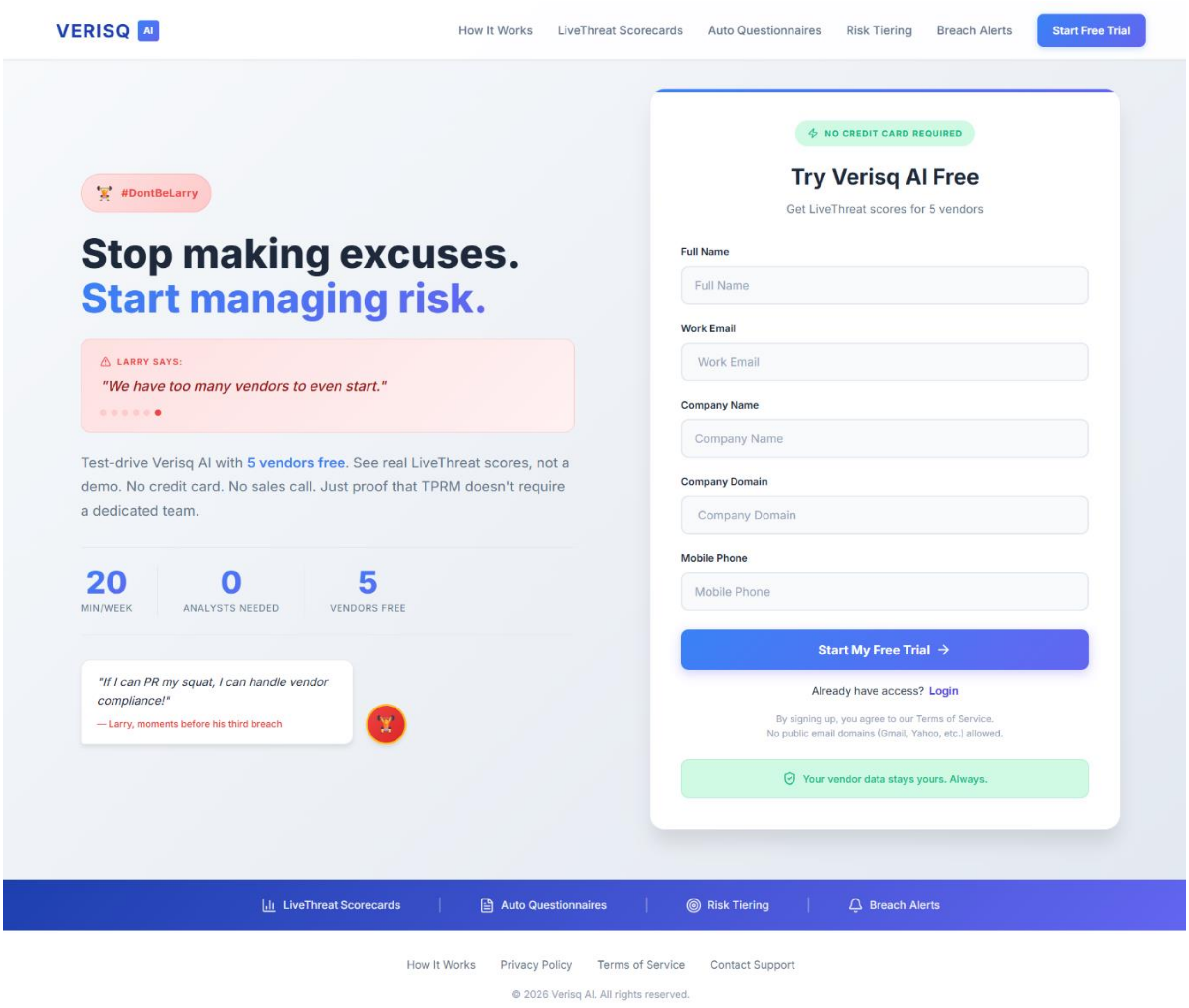
6.2.1 Sitemap

Module	Pages / Routes	Description
Public / Marketing	Landing Page (/), How It Works (/how-it-works)	Introduces VerisqAI, its features and value proposition to prospective users
Authentication	Login / Signup, OTP Verification (/send-code, /otp_verification), Verify (/verify), Trial Request (/request-received), Admin Trial Requests (/admin/trial-requests)	Handles user registration, email/OTP verification and trial access requests
Dashboard Module	Dashboard (/dashboard)	Displays summary widgets: total vendors, risk distribution, recent findings and quick actions
Vendor Management	Vendors list, Vendor Details (/vendor-details)	Lists onboarded vendors and shows detailed vendor profile, risk score, findings and questionnaire status
Questionnaire Module	Questionnaires (/Questionnaires), Questionnaire Welcome (/questionnaire/:token), Assessment (/questionnaire/:token/assessment), Decline (/questionnaire/:token/decline), Declined (/questionnaire/:token/declined)	Manages sending questionnaires to vendors and the vendor-facing flow for completing or declining a questionnaire
Template Builder	Template Builder (/template-builder), Template Details (/template-details)	Allows users to create and edit assessment templates, sections and questions
Risk & Scorecards	Scorecards (/Scorecard), Risk Tiering (/RiskTiering), Breach Alerts (/BreachAlerts)	Displays computed scorecards, risk tier classification and breach/alert notifications
Admin Panel	Admin Dashboard, Users, Vendors, AI Monitoring, Audit Logs	Administrative pages for managing users, vendors, monitoring AI activity and reviewing system audit logs

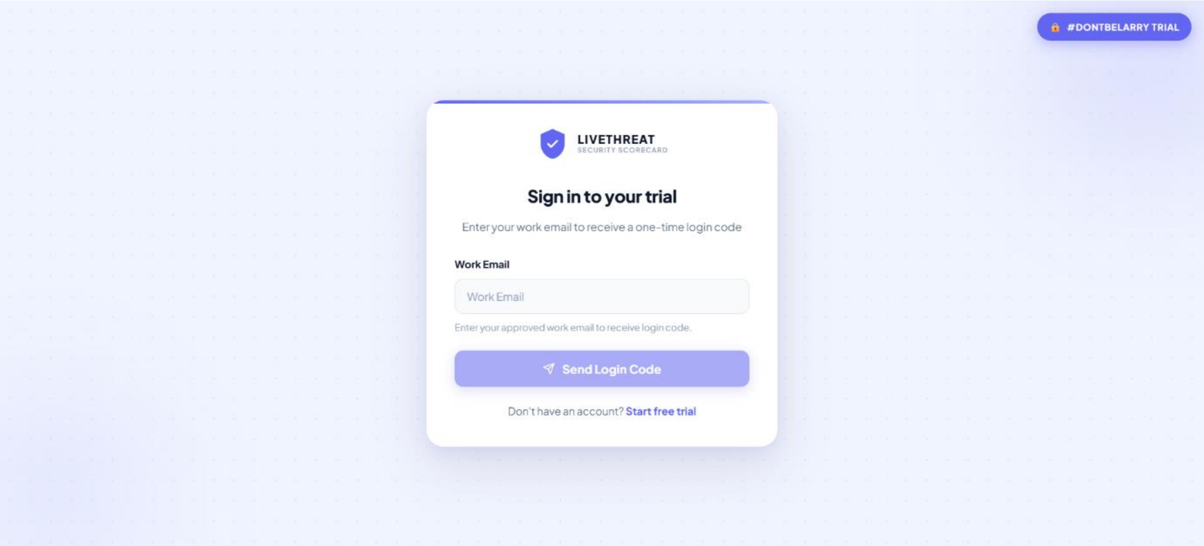
6.2.2 Page Snapshots

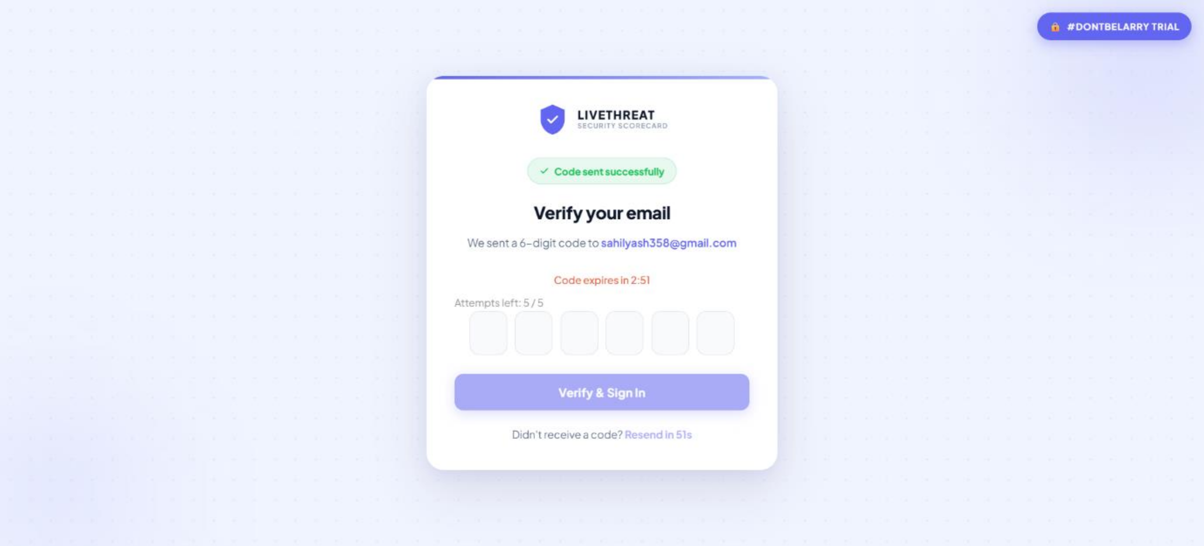
This section presents representative screenshots of the key pages of the VerisqAI application, illustrating the layout, navigation and design consistency across modules.

6.2.2.1 Landing Page

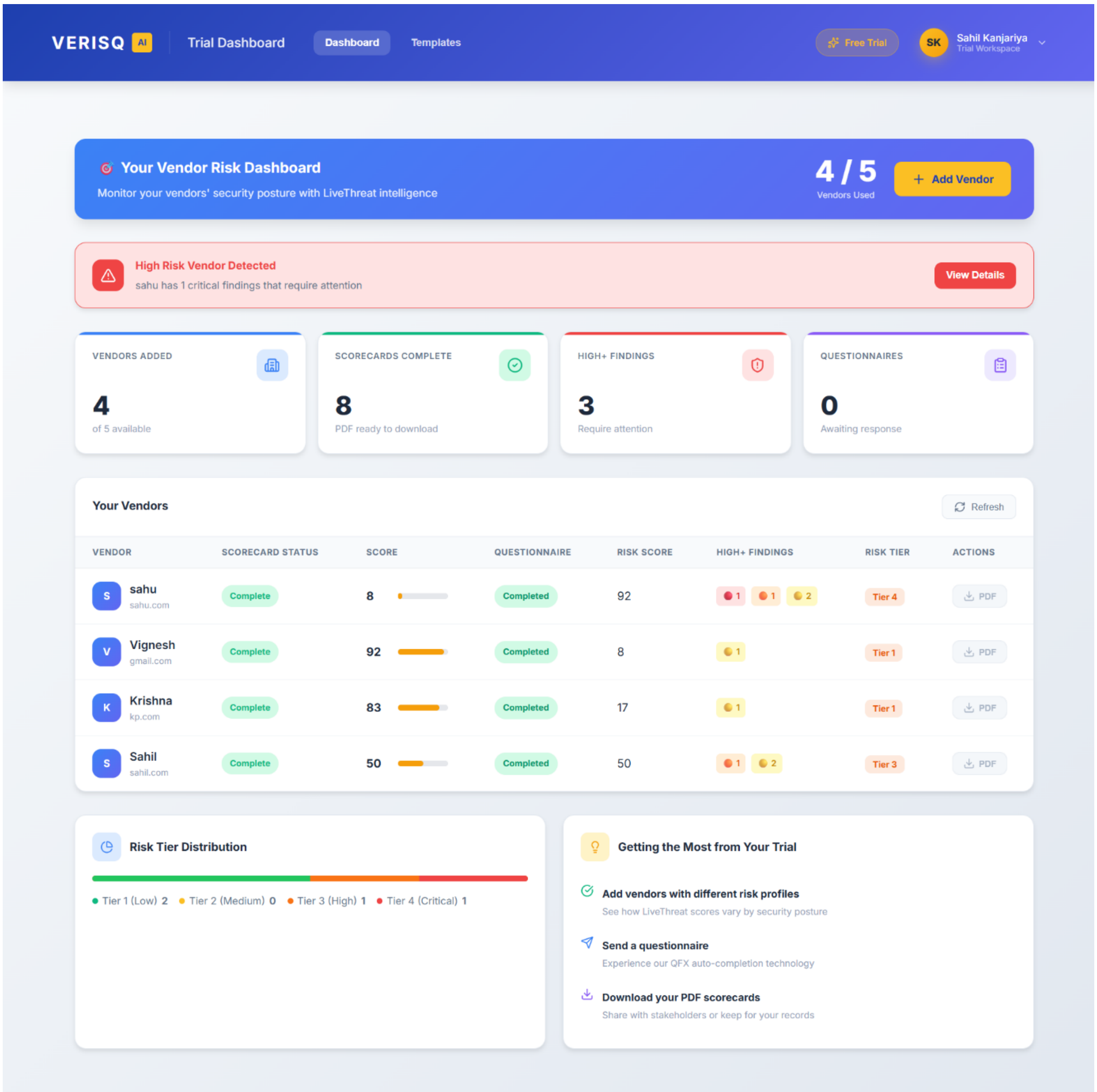


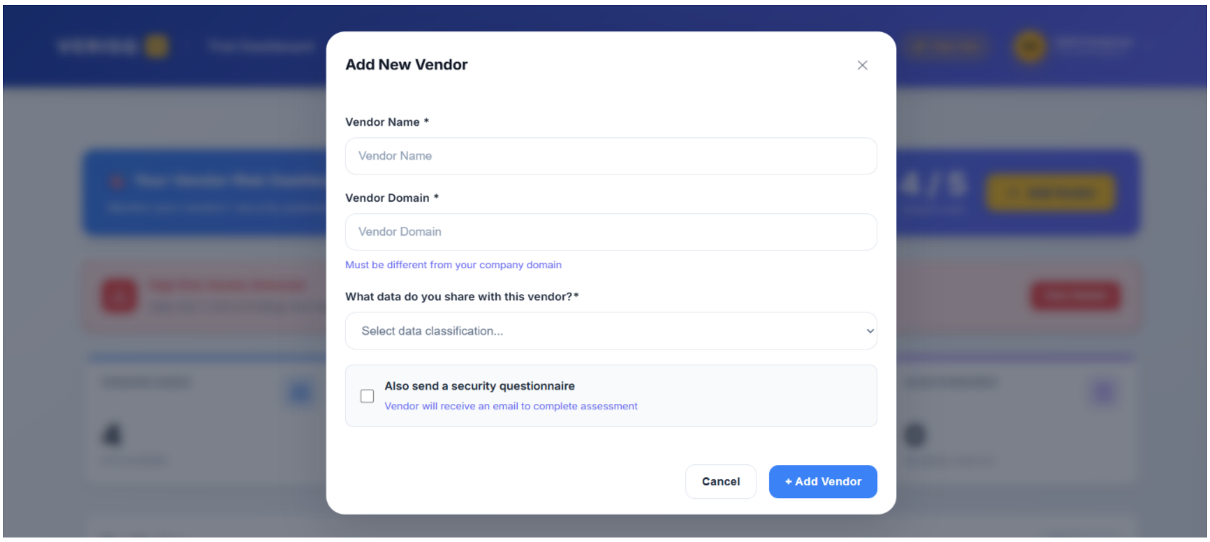
6.2.2.2 Login / OTP Verification Page



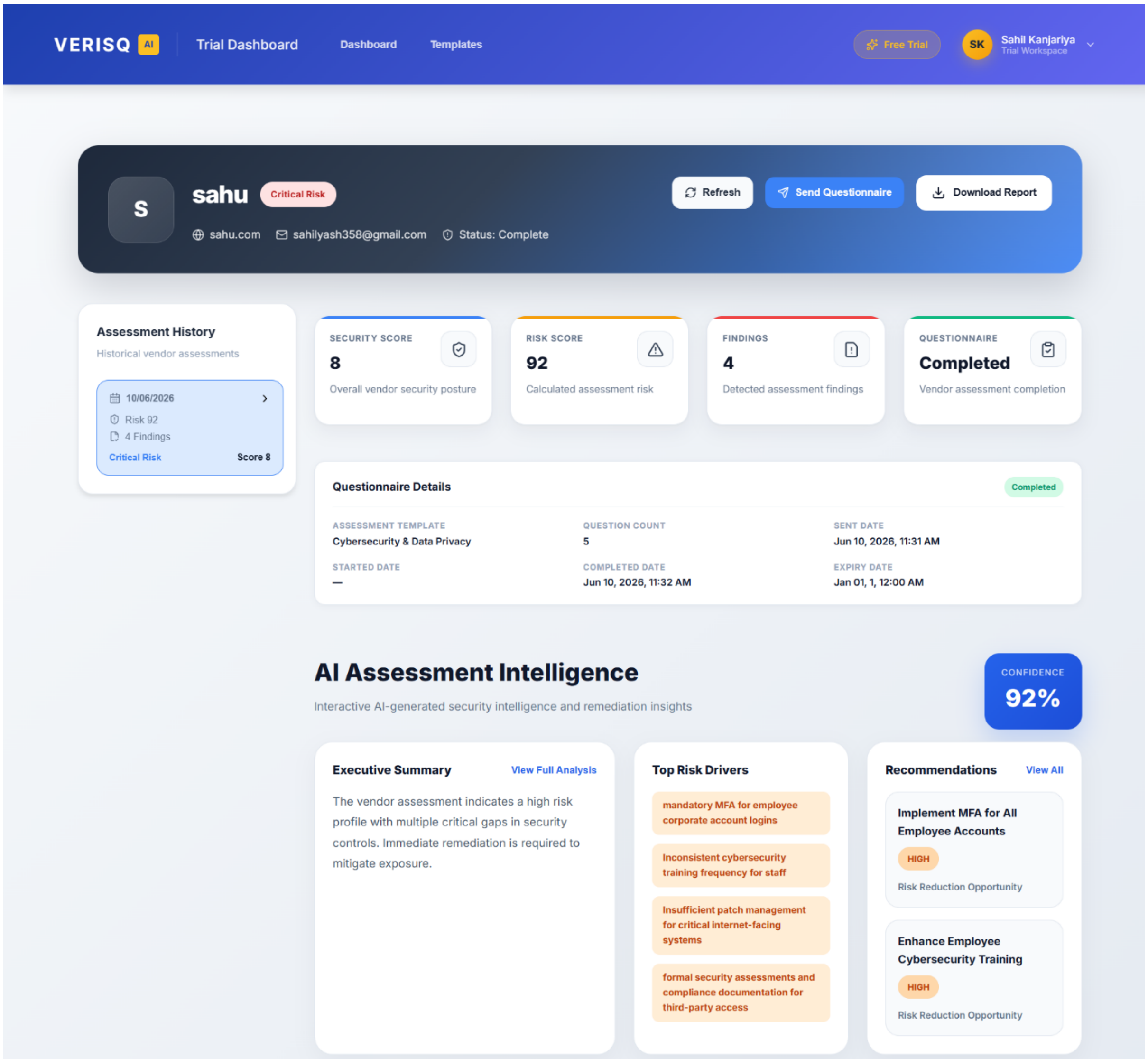


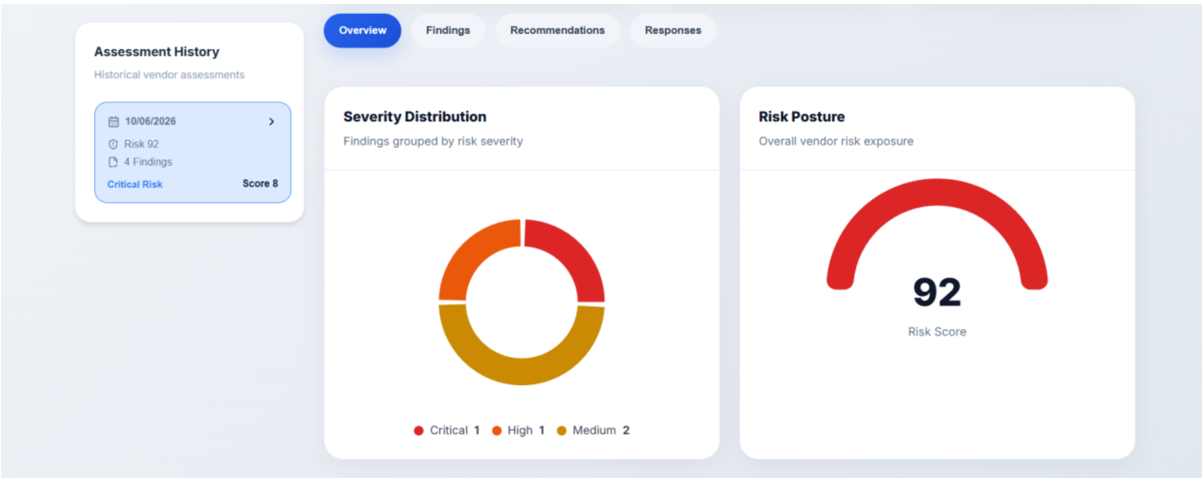
6.2.2.3 Dashboard Page

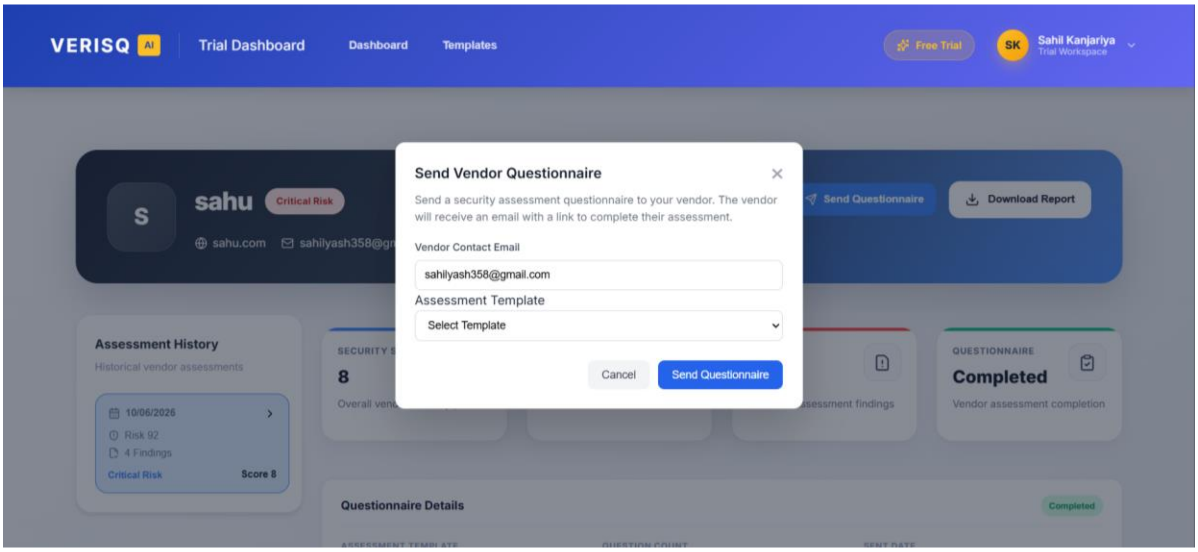




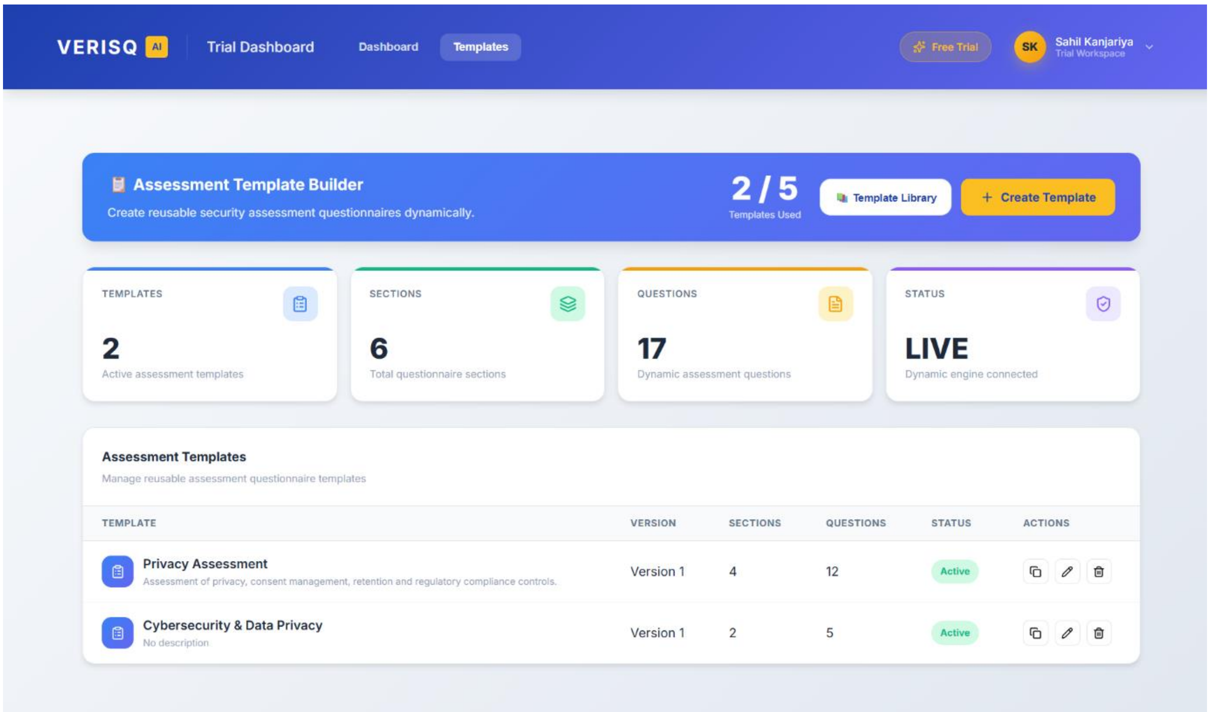
6.2.2.4 Vendor List & Vendor Details Page

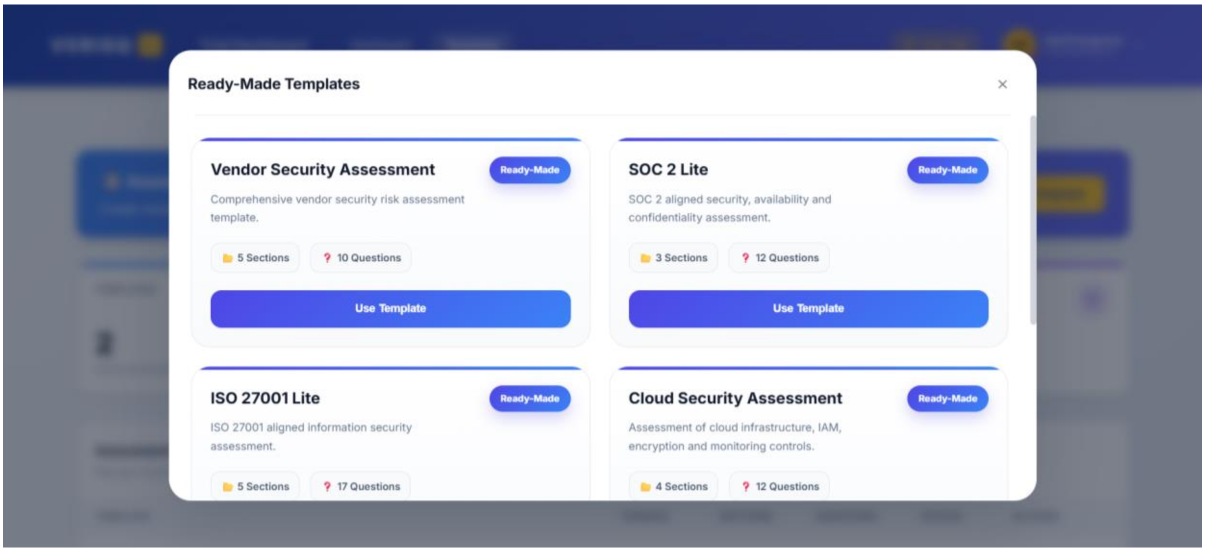


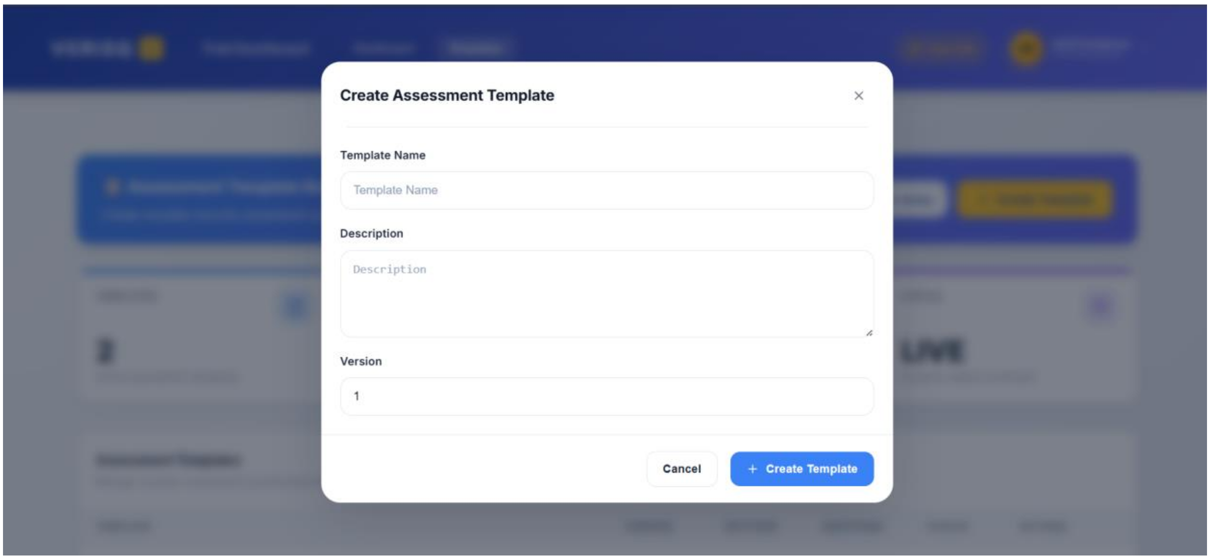


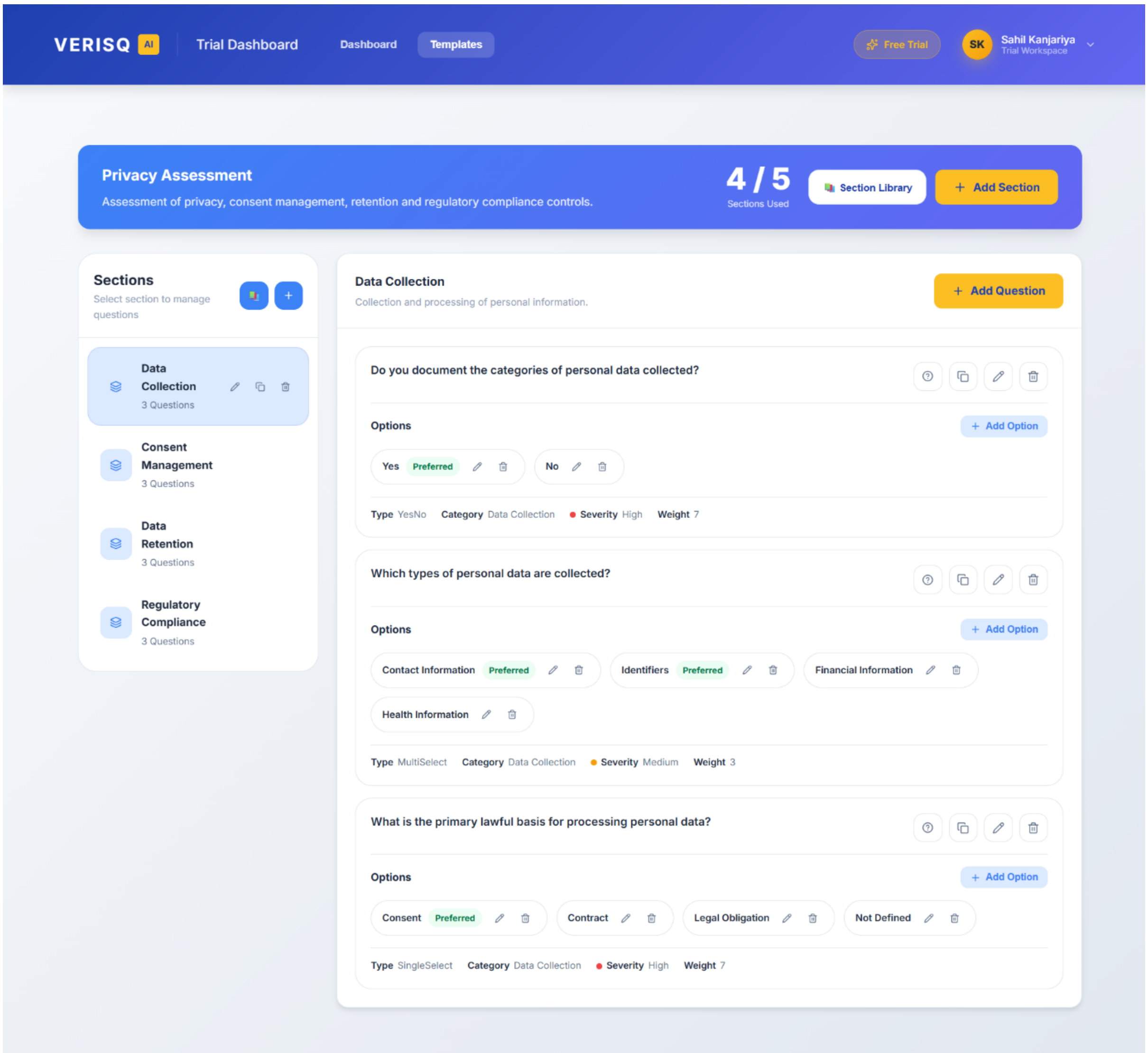


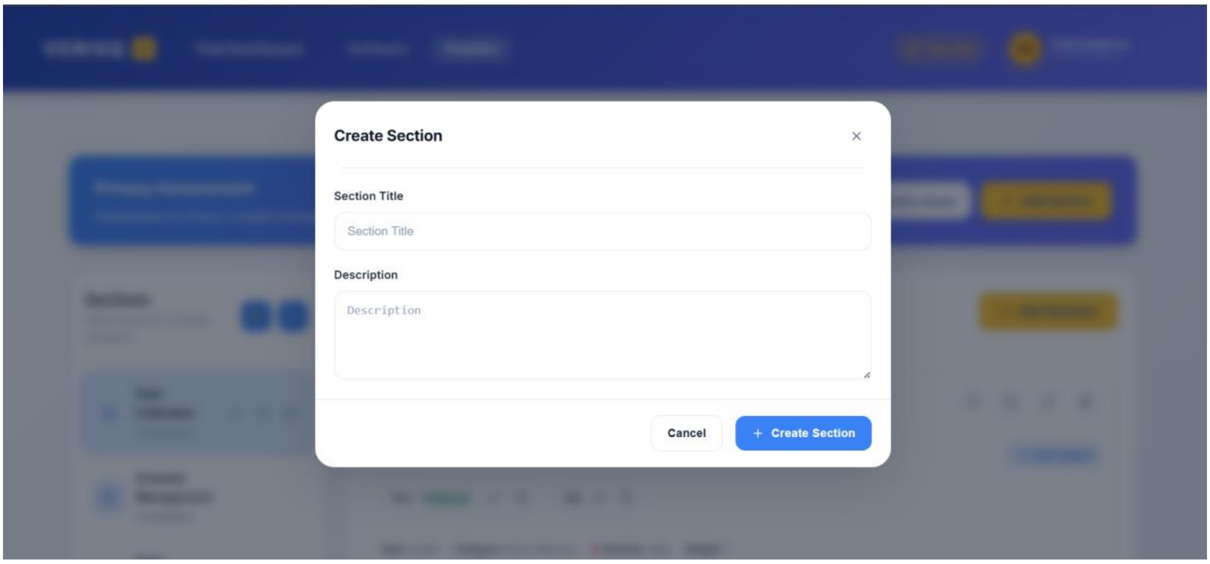
6.2.2.5 Template Builder Page

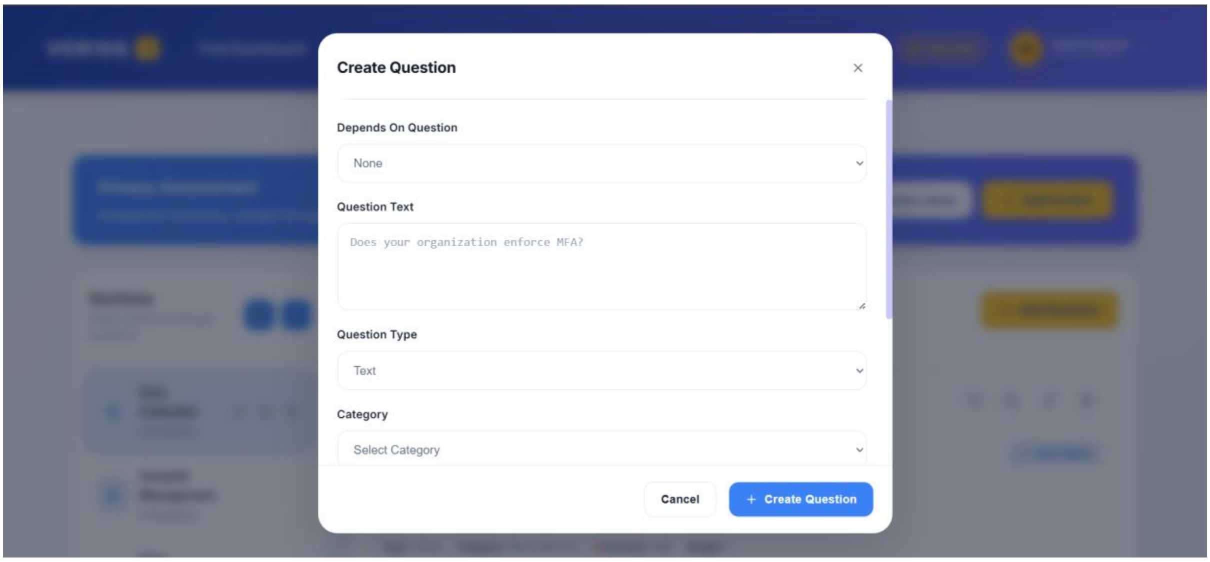



Ready-Made Templates
Vendor Security Assessment
Ready-Made
Comprehensive vendor security risk assessment template.
5 Sections
10 Questions
Use Template
SOC 2 Lite
Ready-Made
SOC 2 aligned security, availability and confidentiality assessment.
3 Sections
12 Questions
Use Template
ISO 27001 Lite
Ready-Made
ISO 27001 aligned information security assessment.
5 Sections
17 Questions
Cloud Security Assessment
Ready-Made
Assessment of cloud infrastructure, IAM, encryption and monitoring controls.
4 Sections
12 Questions

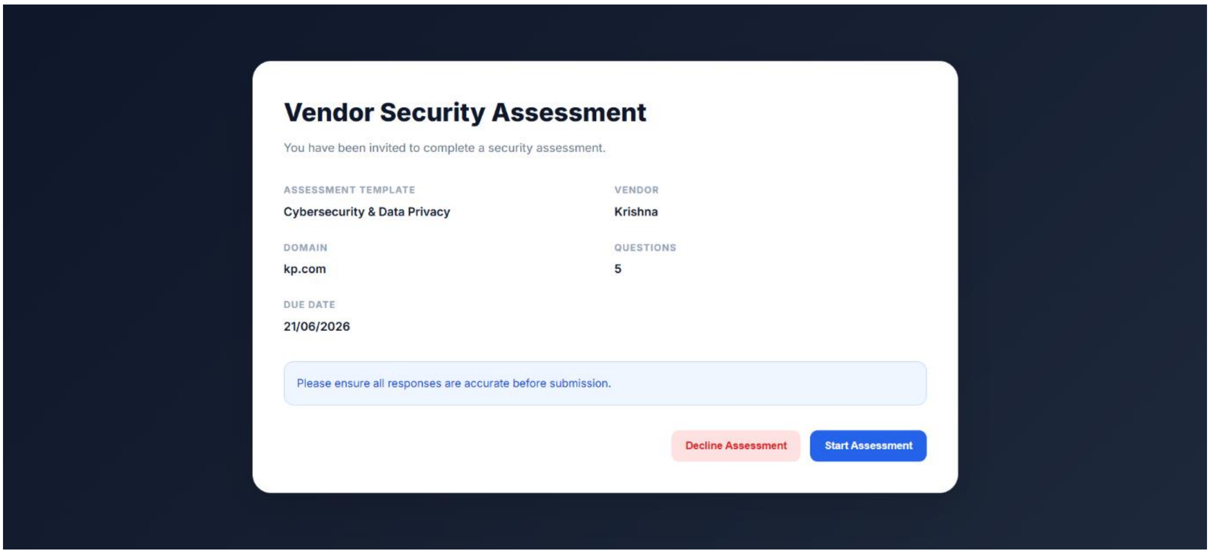

Create Assessment Template
Template Name
Template Name
Description
Description
Version
1
Cancel
Create Template

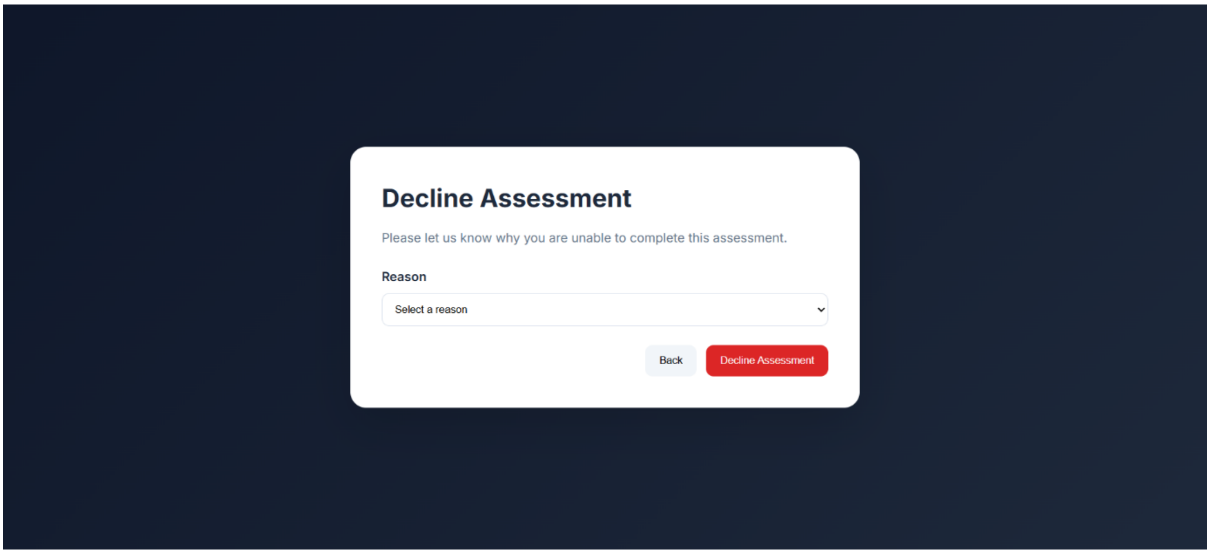

VERISQ AI
Trial Dashboard
Dashboard
Templates
Free Trial
SK
Sahil Kanjariya
Trial Workspace
Privacy Assessment
Assessment of privacy, consent management, retention and regulatory compliance controls.
4 / 5
Sections Used
Section Library
+ Add Section
Sections
Select section to manage questions
Data Collection
3 Questions
Consent Management
3 Questions
Data Retention
3 Questions
Regulatory Compliance
3 Questions
Data Collection
Collection and processing of personal information.
+ Add Question
Do you document the categories of personal data collected?
Options
+ Add Option
Yes Preferred
No
Type YesNo Category Data Collection Severity High Weight 7
Which types of personal data are collected?
Options
+ Add Option
Contact Information Preferred
Identifiers Preferred
Financial Information
Health Information
Type MultiSelect Category Data Collection Severity Medium Weight 3
What is the primary lawful basis for processing personal data?
Options
+ Add Option
Consent Preferred
Contract
Legal Obligation
Not Defined
Type SingleSelect Category Data Collection Severity High Weight 7

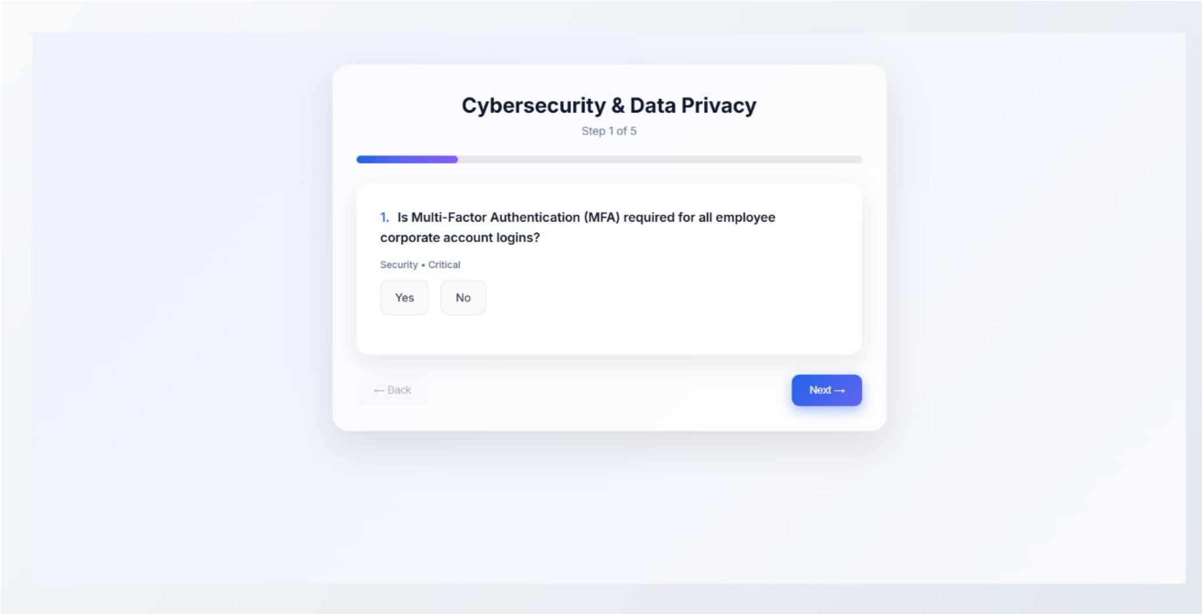

Create Section
Section Title
Section Title
Description
Description
Cancel
+ Create Section



6.2.2.6 Vendor Questionnaire Page

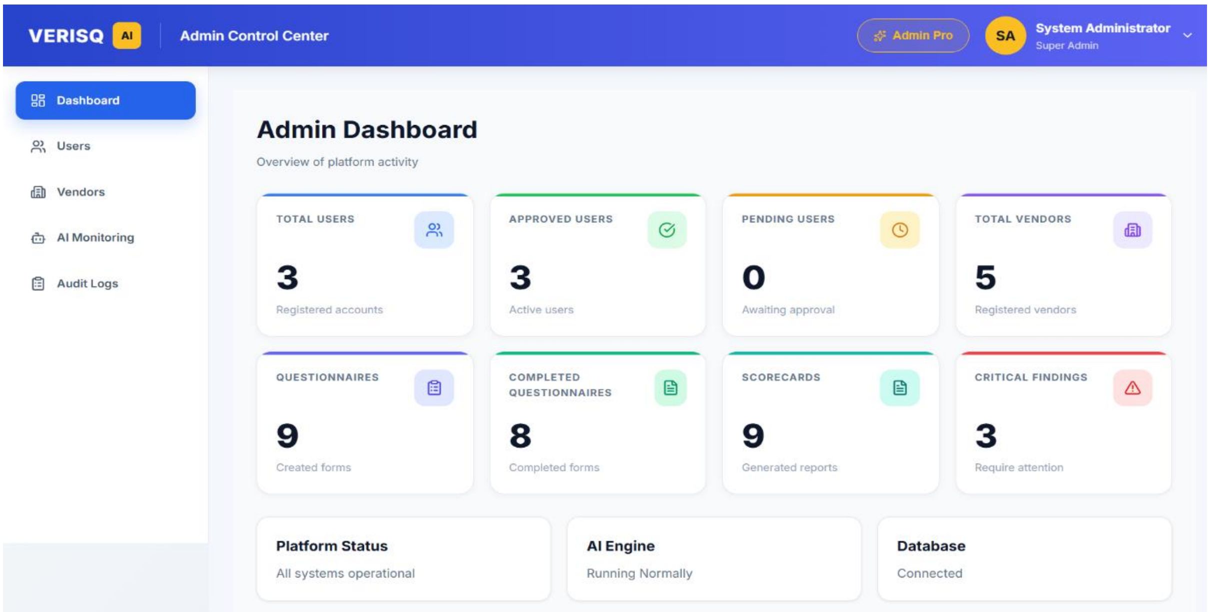


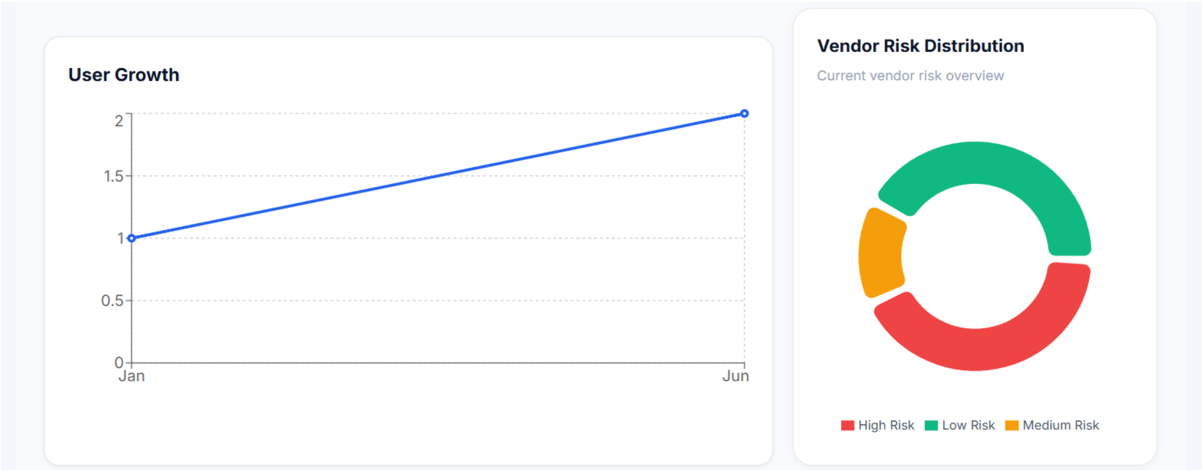




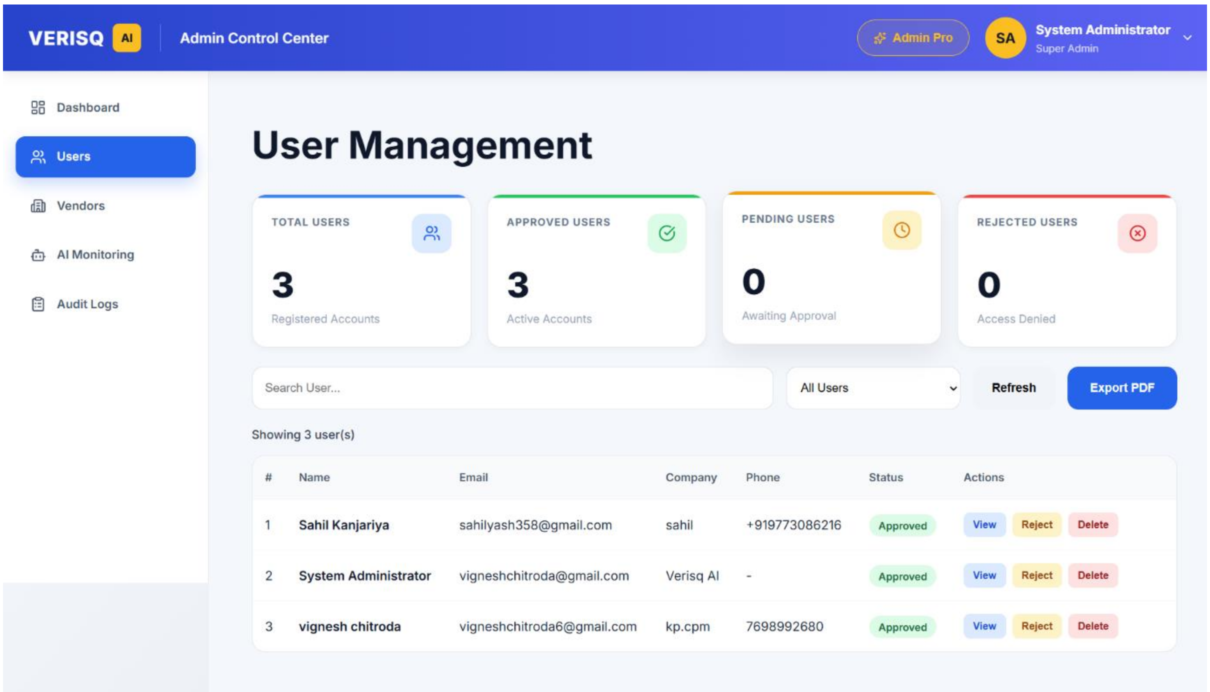
6.2.2.7 Admin Dashboard & AI Monitoring Page

Dashboard

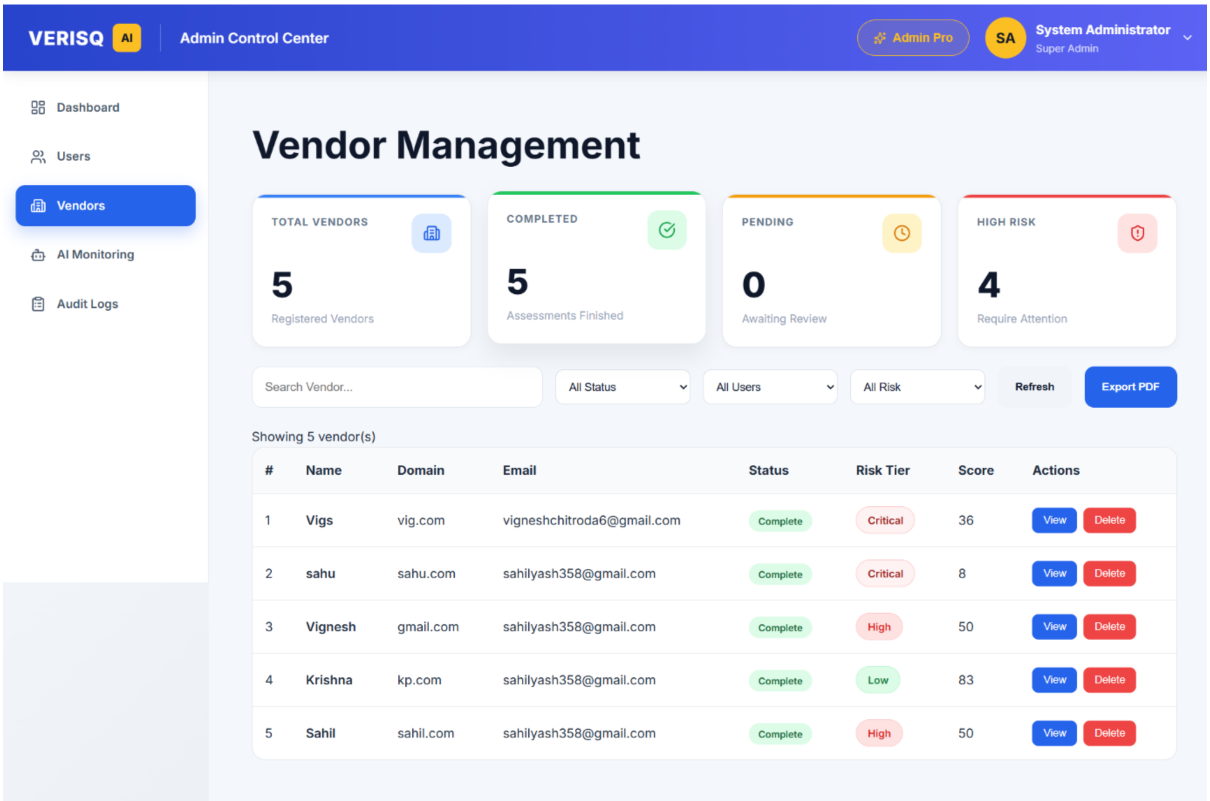




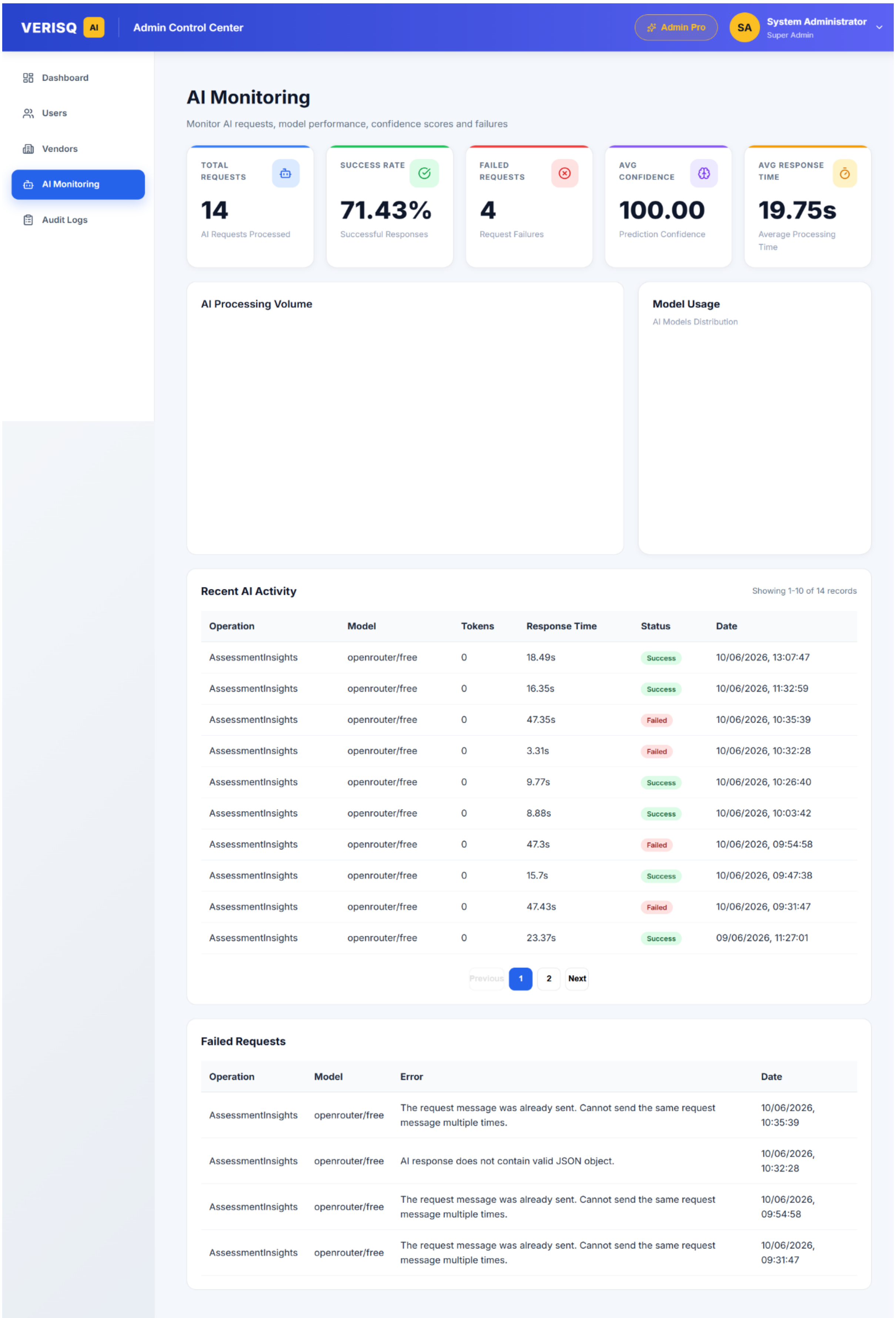
User Management



Vendor Management



Ai Monitoring



Recent AI Activity

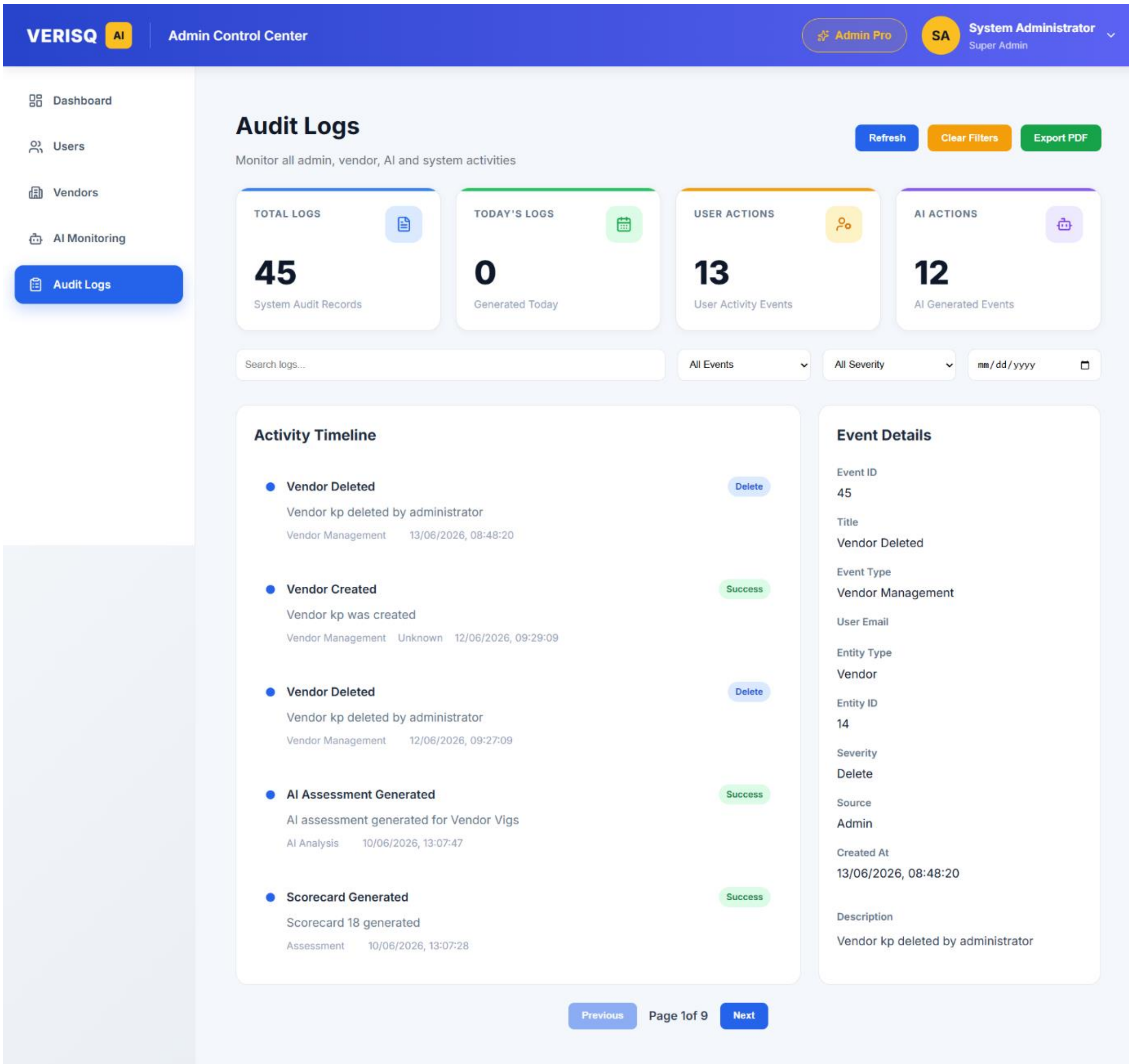
Operation	Model	Tokens	Response Time	Status	Date
AssessmentInsights	openrouter/free	0	18.49s	Success	10/06/2026, 13:07:47
AssessmentInsights	openrouter/free	0	16.35s	Success	10/06/2026, 11:32:59
AssessmentInsights	openrouter/free	0	47.35s	Failed	10/06/2026, 10:35:39
AssessmentInsights	openrouter/free	0	3.31s	Failed	10/06/2026, 10:32:28
AssessmentInsights	openrouter/free	0	9.77s	Success	10/06/2026, 10:26:40
AssessmentInsights	openrouter/free	0	8.88s	Success	10/06/2026, 10:03:42
AssessmentInsights	openrouter/free	0	47.3s	Failed	10/06/2026, 09:54:58
AssessmentInsights	openrouter/free	0	15.7s	Success	10/06/2026, 09:47:38
AssessmentInsights	openrouter/free	0	47.43s	Failed	10/06/2026, 09:31:47
AssessmentInsights	openrouter/free	0	23.37s	Success	09/06/2026, 11:27:01

Failed Requests

Operation	Model	Error	Date
AssessmentInsights	openrouter/free	The request message was already sent. Cannot send the same request message multiple times.	10/06/2026, 10:35:39
AssessmentInsights	openrouter/free	AI response does not contain valid JSON object.	10/06/2026, 10:32:28
AssessmentInsights	openrouter/free	The request message was already sent. Cannot send the same request message multiple times.	10/06/2026, 09:54:58
AssessmentInsights	openrouter/free	The request message was already sent. Cannot send the same request message multiple times.	10/06/2026, 09:31:47


AI Processing Volume
12
9
6
3
0
2026-06-06T00:00:00
2026-06-10T00:00:00
Model Usage
AI Models Distribution
openrouter/free

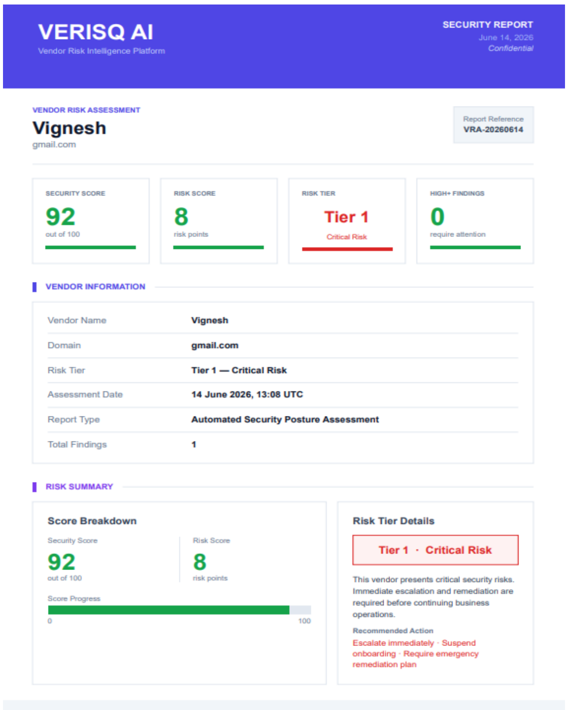
Audit Logs


VERISQ AI
Admin Control Center
Admin Pro
SA
System Administrator
Super Admin
Dashboard
Users
Vendors
AI Monitoring
Audit Logs
Audit Logs
Monitor all admin, vendor, AI and system activities
Refresh
Clear Filters
Export PDF
TOTAL LOGS
45
System Audit Records
TODAY'S LOGS
0
Generated Today
USER ACTIONS
13
User Activity Events
AI ACTIONS
12
AI Generated Events
Search logs...
All Events
All Severity
mm/dd/yyyy
Activity Timeline
Vendor Deleted
Delete
Vendor kp deleted by administrator
Vendor Management 13/06/2026, 08:48:20
Vendor Created
Success
Vendor kp was created
Vendor Management Unknown 12/06/2026, 09:29:09
Vendor Deleted
Delete
Vendor kp deleted by administrator
Vendor Management 12/06/2026, 09:27:09
AI Assessment Generated
Success
AI assessment generated for Vendor Vigs
AI Analysis 10/06/2026, 13:07:47
Scorecard Generated
Success
Scorecard 18 generated
Assessment 10/06/2026, 13:07:28
Event Details
Event ID
45
Title
Vendor Deleted
Event Type
Vendor Management
User Email
Entity Type
Vendor
Entity ID
14
Severity
Delete
Source
Admin
Created At
13/06/2026, 08:48:20
Description
Vendor kp deleted by administrator
Previous
Page 1of 9
Next

6.2.2.8 Admin / User PDF

VerisqAI Audit Logs Report

Event Type	Title	Severity	Source	User	Date
Vendor Management	Vendor Deleted	Delete	Admin		13/06/2026, 08:48:20
Vendor Management	Vendor Created	Success	System	Unknown	12/06/2026, 09:29:09
Vendor Management	Vendor Deleted	Delete	Admin		12/06/2026, 09:27:09
AI Analysis	AI Assessment Generated	Success	AI Engine		10/06/2026, 13:07:47
Assessment	Scorecard Generated	Success	System		10/06/2026, 13:07:28
AI Analysis	AI Assessment Generated	Success	AI Engine		10/06/2026, 11:32:59
Assessment	Scorecard Generated	Success	System		10/06/2026, 11:32:42
AI Analysis	AI Assessment Generated	Success	AI Engine		10/06/2026, 10:35:39
Assessment	Scorecard Generated	Success	System		10/06/2026, 10:34:51
Vendor Management	Vendor Deleted	Delete	Admin		10/06/2026, 10:33:10
AI Analysis	AI Assessment Generated	Success	AI Engine		10/06/2026, 10:32:29
Assessment	Scorecard Generated	Success	System		10/06/2026, 10:32:25
Vendor Management	Vendor Deleted	Delete	Admin		10/06/2026, 10:28:03
AI Analysis	AI Assessment Generated	Success	AI Engine		10/06/2026, 10:26:40
Assessment	Scorecard Generated	Success	System		10/06/2026, 10:26:31
Vendor Management	Vendor Deleted	Delete	Admin		10/06/2026, 10:12:37
AI Analysis	AI Assessment Generated	Success	AI Engine		10/06/2026, 10:03:42
Assessment	Scorecard Generated	Success	System		10/06/2026, 10:03:33
AI Analysis	AI Assessment Generated	Success	AI Engine		10/06/2026, 09:54:58
Assessment	Scorecard Generated	Success	System		10/06/2026, 09:54:11
AI Analysis	AI Assessment Generated	Success	AI Engine		10/06/2026, 09:47:38



6.3 Architecture Design

VerisqAI follows a three-tier (3-tier) architecture, which separates the system into the Presentation Layer, the Application/Business Logic Layer, and the Data Layer. This separation improves maintainability, scalability and security, as each layer can be developed, tested and deployed independently.

6.3.1 Presentation Layer (Client Tier)

The presentation layer is built using React.js (with Vite as the build tool) and is responsible for rendering the user interface and handling user interactions. It communicates with the backend exclusively through RESTful API calls made using Axios.

- Built with React.js, organised into reusable components such as the dashboard, questionnaire builder, template builder, vendor details and admin panels.
- Uses React Router for client-side routing, with separate routes for public pages, authenticated user pages (protected via ProtectedRoute) and admin pages (protected via AdminRoute, restricted to users with the Admin role).
- Uses libraries such as Recharts for data visualisation (risk distribution, scorecards), jsPDF/jspdf-autotable for exporting reports, and lucide-react/react-icons for UI icons.
- User authentication state and role information are stored client-side (JWT token, decoded using jwt-decode) and attached to API requests for authorisation.

6.3.2 Application / Business Logic Layer (Server Tier)

The middle tier is implemented as an ASP.NET Core Web API (built on .NET 8) that exposes RESTful endpoints consumed by the React front-end. This layer contains the core business logic, authentication and AI processing services.

- Controllers (e.g., AuthController, DashboardController, QuestionnaireController, AssessmentTemplateController, AdminVendorsController, AdminAiMonitoringController, AdminAuditLogsController) handle incoming HTTP requests and return JSON responses.
- Authentication and authorisation are handled using ASP.NET Core Identity together with JWT Bearer tokens, with role-based access control distinguishing between regular users and Admin users.
- Service classes encapsulate business logic such as risk score calculation, scorecard generation, questionnaire token generation/validation, and OTP verification for vendor email confirmation.
- An AI module (under the AI folder) integrates with an external AI service to analyse vendor questionnaire responses, generate AI-based risk insights (AiAssessmentInsight) and remediation recommendations (AiRecommendation), and logs all AI activity via AiProcessingLog and AiExecutionAudit for monitoring through the Admin AI Monitoring dashboard.
- QuestPDF is used on the server side to generate downloadable PDF reports such as vendor scorecards.
- All significant actions (vendor onboarding, questionnaire sent/completed, AI processing, admin actions) are recorded in the AuditLog table for compliance and traceability.

6.3.3 Data Layer

The data layer consists of a Microsoft SQL Server database accessed through Entity Framework Core using a Code-First approach with migrations. ApplicationDbContext serves as the single entry point for all database operations, mapping the entities described in Section 6.1 to corresponding tables (Vendors, Questionnaires, AssessmentTemplates, AssessmentSections, AssessmentQuestions, QuestionnaireResponses, Scorecards, Findings, AuditLogs, AI-related tables, and ASP.NET Identity tables).

- EF Core Migrations maintain the database schema and allow incremental, version-controlled changes (e.g., AddCoreEntities, AddDynamicAssessmentEngine, AddAiAssessmentInfrastructure, AddAuditLogs).
- Seed data classes pre-populate default assessment templates, sections and questions so that new organisations can immediately start sending vendor questionnaires.
- Data integrity is enforced through primary/foreign key constraints linking Vendors, Questionnaires, Scorecards, Findings and AssessmentTemplates.

6.3.4 Three-Tier Architecture Diagram

Figure 6.11 illustrates the flow of data between the three tiers: a user interacts with the React-based presentation layer in the browser, which sends HTTPS/REST requests to the ASP.NET Core Web API (application layer). The application layer processes business logic, communicates with the AI engine for risk analysis, and performs CRUD operations on the SQL Server database (data layer) via Entity Framework Core. Responses flow back through the same path to update the user interface.

Tier	Technology / Components	Responsibility
Presentation Tier (Client)	React.js, Vite, React Router, Axios, Recharts, jsPDF	Renders UI, handles navigation and user input, sends REST API requests, displays dashboards, scorecards and reports
Application Tier (Server)	ASP.NET Core 8 Web API, Identity + JWT, AI integration module, QuestPDF	Implements business logic, authentication/authorization, risk scoring, AI-based insight generation, PDF report generation, audit logging
Data Tier (Database)	Microsoft SQL Server, Entity Framework Core (Code-First, Migrations)	Persistent storage of vendors, questionnaires, templates, scorecards, findings, AI insights and audit logs

This three-tier separation allows the front-end to be deployed independently (e.g., on Netlify) from the back-end API and database, supports horizontal scaling of the application layer, and ensures that sensitive data access and AI processing logic remain secured within the server tier, away from direct client access.

7. Testing

7.1 Unit Testing

Unit testing for VerisqAI was performed on individual service and controller methods to verify that each component performs its intended function correctly in isolation. The following key unit tests were conducted:

Test Case	Description	Expected	Result
Auth_Login_ValidCredentials	User logs in with correct email/password	JWT token returned	Pass
Auth_Login_InvalidPassword	User logs in with wrong password	401 Unauthorized	Pass
Auth_OTP_ExpiredToken	OTP verification with expired token	400 Bad Request	Pass
Vendor_Add_ValidData	Add vendor with all required fields	Vendor created with status Queued	Pass
Vendor_Add_DuplicateName	Add vendor with same name twice	Vendor added (no unique constraint)	Pass
Template_Create_ValidTemplate	Create template with sections and questions	Template saved with hierarchy	Pass
Questionnaire_Send_ValidVendor	Send questionnaire to existing vendor	Token generated, status Sent	Pass
Questionnaire_Submit_ValidToken	Submit questionnaire with valid token	Responses saved, status Completed	Pass
Questionnaire_Submit_ExpiredToken	Submit questionnaire with expired token	400 Bad Request	Pass
Scoring_CalculateScore_AllCorrect	All preferred answers selected	Score = 100, Tier = 1	Pass

Scoring_CalculateScore_AllWrong	All non-preferred answers selected	Score = 0, Tier = 4	Pass
Scoring_FindingGeneration	Critical question answered incorrectly	Critical finding created	Pass
AI_ParseResponse_ValidJSON	AI returns well-formed JSON	Insight and recommendations stored	Pass
AI_ParseResponse_InvalidJSON	AI returns malformed response	Error handled, no crash	Pass
PDF_Generate_CompletedScorecard	Generate PDF for completed vendor	PDF file returned	Pass

7.2 Integration Testing

Integration testing was performed to verify that the various modules of VerisqAI work correctly together as a complete system. The following integration test scenarios were executed:

Integration Test	Flow Tested	Expected	Result
End-to-End Vendor Assessment	User login → Add vendor → Create template → Send questionnaire → Vendor submits → Scorecard generated → AI analysis → PDF download	Complete scorecard with AI insights and PDF	Pass
Auth + Vendor Management	Login with JWT → Access protected vendor endpoints → Unauthorized access attempt without token	Correct 401 on unauthorized access	Pass
Template → Questionnaire Link	Create template → Send questionnaire using that template → Verify correct questions appear in vendor form	Questions match template definition	Pass
Scoring → AI Integration	Submit questionnaire → Score calculated → AI analysis triggered → Insights stored in same scorecard	Scorecard has both scores and AI insights	Pass

Admin Dashboard Data	Multiple users add vendors → Admin accesses dashboard → Correct aggregated statistics shown	Admin sees all vendor risk distribution	Pass
Audit Log Generation	User performs login, adds vendor, sends questionnaire → Audit logs created for each action	Three audit log entries created	Pass
Questionnaire Expiry	Send questionnaire with past expiry → Vendor attempts to access → Form is rejected	Token rejected with appropriate message	Pass
Concurrent Questionnaire Submissions	Two vendors submit at the same time → Both scorecards generated independently	No data mixing; correct scorecards	Pass

8. Future Enhancement

The following enhancements are planned for future phases of VerisqAI development:

- Template Preview Mode (Phase 2B): Allow risk analysts to preview exactly how a questionnaire will appear to the vendor before dispatching it, enabling quality review of question flow and formatting.
- Bulk Vendor Import: Support CSV/Excel-based bulk vendor onboarding so that organizations with large vendor portfolios can import hundreds of vendors at once without manual data entry.
- Multi-Language Questionnaire Support: Add internationalization support so questionnaires can be presented in vendors' preferred languages, improving response rates for global vendor ecosystems.
- Mobile Application: Develop native iOS and Android applications for risk analysts to monitor vendor statuses and receive push notifications about questionnaire completion.
- Vendor Self-Registration Portal: Allow vendors to self-register on the platform so they can proactively submit compliance documentation without waiting for a manual invitation.
- Third-Party Security Ratings Integration: Integrate with external security rating services (e.g., SecurityScorecard, BitSight) to enrich vendor risk profiles with externally sourced data.
- Scheduled Re-Assessment: Implement automated questionnaire scheduling to trigger periodic re-assessments for high-risk vendors at configurable intervals (quarterly, annually).

9. Glossary

Term	Definition
AI (Artificial Intelligence)	Technology that enables machines to simulate human intelligence for tasks such as analysis, decision-making, and natural language understanding.
API (Application Programming Interface)	A set of rules that allows different software applications to communicate with each other over a network.
ASP.NET Core	A cross-platform, high-performance, open-source framework for building modern, cloud-based web applications by Microsoft.
Audit Log	A chronological record of system events and user actions used for security monitoring and compliance purposes.
CRC (Class Responsibility Collaborator)	A UML design technique that identifies classes, their responsibilities, and which other classes they collaborate with.
CRUD	Create, Read, Update, Delete – the four basic operations performed on database records.
DTO (Data Transfer Object)	An object used to transfer data between software layers without exposing internal entity structures.
EF Core (Entity Framework Core)	An open-source ORM (Object-Relational Mapper) for .NET that allows developers to work with databases using .NET objects.
Finding	A specific security risk or compliance gap identified during a vendor assessment, categorized by severity (Critical, High, Medium, Low).
JWT (JSON Web Token)	A compact, URL-safe token format used for securely transmitting authentication information between parties.
LLM (Large Language Model)	A type of AI model trained on large amounts of text data capable of generating human-like text and performing analysis tasks.
ORM (Object-Relational Mapper)	A programming technique that converts data between incompatible type systems (objects and relational databases).
OTP (One-Time Password)	A password that is valid for only one login session or transaction, used as a second factor in authentication.

OpenRouter	An AI API aggregation service that provides access to multiple LLM providers through a unified API interface.
QuestPDF	An open-source .NET library for generating professional PDF documents using a fluent API.
React	A JavaScript library for building user interfaces, maintained by Meta (Facebook) and widely used for SPAs.
Risk Score	A numeric value (0-100) calculated based on vendor assessment responses indicating the overall risk level of a vendor.
Risk Tier	A categorical risk classification (1-4) assigned to a vendor based on their risk score: Tier 1 (Low), Tier 2 (Medium), Tier 3 (High), Tier 4 (Critical).
Scorecard	A comprehensive record containing a vendor's risk score, risk tier, findings, and AI-generated insights from a completed assessment.
SRS (Software Requirements Specification)	A document that describes the behavior, functionalities, and constraints of a software system.
TOTP (Time-based One-Time Password)	An OTP algorithm that generates a password valid only for a short time window, typically 30 seconds.
Vendor	A third-party organization or supplier whose security practices and compliance posture is being assessed.
VRM (Vendor Risk Management)	The process of identifying, assessing, and mitigating risks associated with third-party vendors and suppliers.

10. References

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6. Claude (Anthropic) — Used for documentation writing and project-related guidance. https://claude.ai
7. GitHub — Referred to an open-source e-commerce project repository for understanding project structure and implementation approach. https://github.com
8. Stack Overflow – Used for solving errors. https://stackoverflow.com/questions